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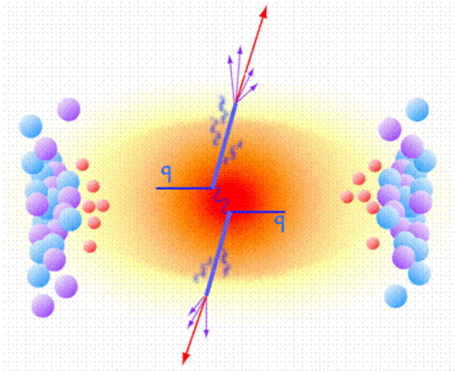

Medium Modified Perturbative QCD

Calculation of Splitting Functions

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Chapter 3

Medium Propagation

In the last chapter, we saw one of the evolution equations for particle propagation in the vacuum. This means that what we saw were elementary reactions or light nuclear reactions whose offspring propagates through the vacuum so that it can be detected. Now, with the LHC program and even with RHIC, this situation is no longer real. Because reactions are with heavy nuclei, like Pb-Pb collisions, the hadronic projectiles have now spatial extension, which is proportional to $A^{1/3}$. So, there is a need for additional physical input so one can extend the application of QCD to the description of this kind of processes with high-$p_\perp$ parton cross sections. This discussion leads to some important effects which have to be taken into account:

- First, the density of the incoming parton distribution increases for a given x and Q^2 which means that non-linear modifications of the QCD evolution equations are now relevant;
- The QCD evolution of the initial hadronic wave functions may be altered due the presence of a spatially extended medium through which each of the intervenients of the reaction propagates. This will result in multiple scattering phenomena and energy loss;
- Also, the presence of spatially extended matter can affect the fragmentation and hadronization of hard partons produced in nucleus-nucleus (AA) collisions. It is expected to affect essentially all hight-$p_\perp$ hadronic observables in heavy ion collisions at collider energies.

The main motivation for the study of these observables is that the degree of nuclear modification may allow for a detailed characterization of the hot and dense matter produced in the collision, the Quark Gluon Plasma [35,36]. This new form of matter consists of an extended volume of deconfined chirally-symmetric quarks and gluons. Since the lifetime of this medium is so small, of the order of its own transverse size, only indirect probes are available to characterize its thermodynamical and transport properties.

But how do we know that this new state of matter was formed? One of the first proposals to signal its formation was Jet Quenching [37,38], a collection of parton energy loss phenomena inside the medium that leads to the attenuation or disappearance of the spray of hadrons resulting from the fragmentation of a parton. In figure 3.1, we have a hard scattering of two quarks, where one goes out directly to the vacuum and hadronize; meanwhile the other goes through the medium created, suffers energy loss by medium-induced gluon radiation, and then fragments outside in a less energetic jet, a quenched jet.

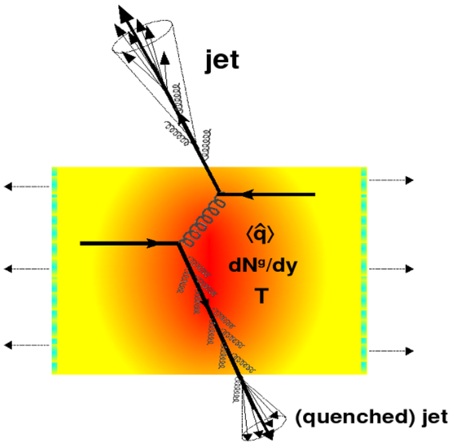


Figure 3.1: Jet Quenching in a AA collision

The usual way of showing the energy loss is through the comparison between the results of a given observable measured in a AA collision (Φ_{AA}) and the results of the same variable but now in a proton-proton (pp) collision (Ψ_{pp}). This ratio called nuclear modification factor [38, 39], is a function of the CM energy ($\sqrt{s}$), transverse momentum ($p_\perp$), rapidity (y), particle mass (m) and impact parameter (b):

$$R_{AA}(\sqrt{s},\, p_\perp,\, y,\, m\, b) = \frac{\textit{hot/dense QCD medium}}{\textit{QCD vacuum}} = \frac{\Phi_{AA}(\sqrt{s},\, p_\perp,\, y,\, m\, b)}{\Phi_{pp}(\sqrt{s},\, p_\perp,\, y,\, m\, b)} \tag{3.1}$$

Any enhancement ($R_{AA} > 1$) or suppression ($R_{AA} < 1$) of this factor can be linked to the properties of the QGP. A quantitive version of this equation can be the ration of the momentum spectrum from AA scatterings and suitably normalized momentum spectrum from pp scatterings:

$$R^h_{AA}(p_\perp) = \frac{(dN^h_{AA}/dp_\perp)}{N_{binary}(dN^h_{pp}/d_\perp)} \tag{3.2}$$

where h denotes the observed hadron species and N_{binary} is the average number of hard binary collisions at the specified impact parameter.

In a general way, the total energy loss of a particle transversing a medium is the sum of collisional and radiative terms:

$$\Delta E = \Delta E_{coll} + \Delta E_{rad} \tag{3.3}$$

The collisional energy loss is made through elastic scatterings with the medium constituents and dominates at low energies. At higher energies, the main mechanism responsible for energy loss is the inelastic scattering. These processes are responsible for the radiative terms from equation (3.3).

The energy lost by a particle in a medium depend on the characteristics of the particle itself and on the plasma properties. So, a mixture of these two can give variables very useful to describe the propagation.

- The mean free path

$$\lambda = (\rho\sigma)^{-1} \tag{3.4}$$

where ρ is the the medium density and σ the cross section of the particle-medium interaction;

- The opacity

$$N = \frac{L}{\lambda} \tag{3.5}$$

where L is the medium thickness;

- The Debye mass, m_D which characterizes the typical momentum exchange with the medium, $\langle q_\perp^2 \rangle$;
- the diffusion constant D, that encodes the dynamics of heavy non-relativistic particles crossing the plasma;
- The transport coefficient $\hat{q}$ that measures the *scattering power* of the medium

This last parameter can be obtained, for example, via medium-modified fragmentation functions written as $D^{med}_{parton\to hadron}(z, Q^2)$, with z being the fraction of the jet energy carried by a hadron. This modified function can be schematically written as:

$$D^{med}_{i\to h}(z, \hat{q}, L) \simeq P(z, \hat{q}, L) \otimes D^{vac}_{i\to h} \tag{3.6}$$

where the correction $P(z, \hat{q}, L)$ can be connected to the QCD splitting functions. These must be modified to take into account medium-induced gluon radiation and for that, there exists at least four major phenomenological approaches which have been developed. Their identification is often made with the initials of their authors: BDMPS-Z/ASW [38–40,43], GLV [38,39,41], Higher Twist (HT) [38,39,42] and a finite temperature field theory approach called AMY [38,39,43]. All these approaches have to handle with many difficulties:

- There is no direct measurement of the transversing parton but (in the best case) only of the final state hadrons issuing from its fragmentation;
- The partons crossing the medium can be produced at any initial point within the fireball;
- Because the medium is evolving, the characteristics of the plasma are both time and position-dependent: $m_D(\mathbf{r}, t)$, $\hat{q}(\mathbf{r}, t)$, ...;
- The finite size of the QGP and associated energy loss fluctuations have to be taken into account;
- Perturbation theory cannot be directly applied to computation of cross sections because: the dispersion relation of a particle is no longer that in the vacuum; the scattering potentials are screened; extra divergences appear due to frequent soft exchanges.

With all these, to make calculations possible, one have to assume that factorization like (1.20) and (3.6) is still possible, although not yet fully proved [44]. So, all efforts of these approaches are concentrated on the calculation of the medium-modified parton fragmentation function into hadrons:

$$D^{vac}_{c\to h}(z) \to D^{med}_{c\to h}(z', \hat{q}) \tag{3.7}$$

In all of them, the final hadronization of the hard parton is assumed to occur in the vacuum after the parton, with degraded energy $z' < z$, has escaped from the system, like pictured in figure 3.2.

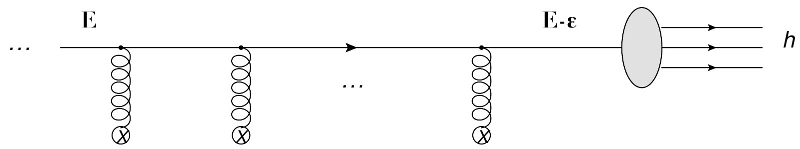


Figure 3.2: Schematic representation of the hadronization process in the medium.

So, the main goal of these approaches is to compute the change in the gluon radiation spectrum from a hard parton due to the presence of the medium. It is always assumed that the forward energy of the jet far exceeds the medium scale ($E \gg \mu$). Also, the temperature from the plasma is considered to be high enough so that perturbation theory can be applied. The differences between the models settles in their assumptions about:

- The relation between parton energy E/ virtuality Q^2 and typical momentum m_D/ spatial extent L of the medium;
- The space-time profile of the medium and its evolution;
- Resumming of diagrams needed to describe the process;
- How to treat the scattering centres.

In the next sub-section, it is possible to find a brief description of them.

3.1 Jet Quenching Models

3.1.1 BDMPS-Z/ASW

The first study of radiative energy loss was conducted by Baier, Dokshitzer, Mueller, Peigné, Schiff (BDMPS) and Zakharov. Nowadays, the most widespread variant of this approach was developed by Armesto, Salgado and Wiedemann (ASW).

This formalism assumes a model for the medium as an assembly of Debye screened heavy scattering centres which are well separated in the sense that the colour screening length of the medium, $1/\mu$, is much smaller than the mean free path of a jet, λ: $\lambda \gg 1/\mu$. A hard parton, almost on-shell, crossing this medium, interacts with various scattering centres through multiple scatterings, and splits into an outgoing parton as well as a radiated gluon with transverse momentum $k_\perp$ and energy ω, which also scatter multiply in the medium (see figure 3.3).

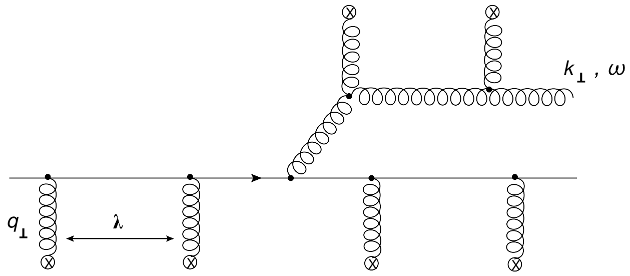


Figure 3.3: Typical gluon radiation diagram in the BDMPS approach

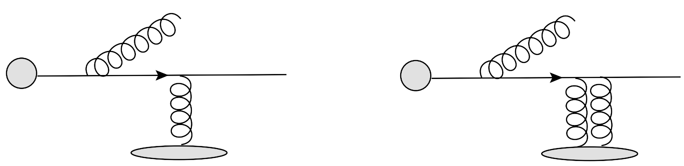
The propagation of the transversing parton and radiated gluon may be expressed in terms of Green's functions which are obtained by a path integral over fields. The final outcome of the approach is a complex analytical expression for the radiated gluon energy distribution as a function of the transport coefficient $\hat{q}$. Because the gluon is considered in the soft limit ($\omega \to 0$), multiple gluon emissions are required for a substantial amount of energy loss. Each emission is assumed to be independent and probabilistic (follows a Poisson distribution). This probability, that the propagating parton loses a fraction of energy ϵ due to n gluon emissions, is encoded in quenching weights or Sudakov form factors $P_\epsilon(\epsilon, \hat{q})$. These include the splitting functions and are used to compute a medium modified fragmentation function:

$$D^{med}_{i \to h}(z', Q^2) = P_\epsilon(\epsilon, \hat{q}) \otimes D^{vac}_{i \to h}(z, Q^2) \tag{3.8}$$

Because the medium is considered a static QGP, no elastic energy loss is vanishing in this scheme. Also, any flavour changing in the medium is neglected.

3.1.2 GLV

The Gyulassy-Lévai-Vitev approach calculate the parton energy loss in a dense deconfined medium, consisting of almost static scattering centres producing a screened Coulomb potential, such as the BDMPS-Z/ASW model. Although both formalisms are equivalent, the GLV starts from the single-hard radiation spectrum (see figure 3.4(a)) rather than the multiple-soft bremsstrahlung. Then, the spectrum is expanded to account for gluon emission from multiple scatterings (see figure 3.4(b)). Each scattering is assumed to be independent and probabilistic like in the previous scheme.



(a) (b)

Figure 3.4: Example of a type of diagrams present in the GLV model

3.1.3 Higher Twist (HT)

The origin of the HT approximation scheme lies in the calculation of medium enhanced higher twist corrections to the total cross section in DIS off large nuclei. This allows one to compute multiple Feynman diagrams, such as those in figure 3.5, assuming the hierarchy scales $E \gg Q \gg \mu$. The process in figure 3.5 is the rescattering of a quark with gluon radiation. It is represented the diagram with its conjugate where the dashed line is one possible cut.

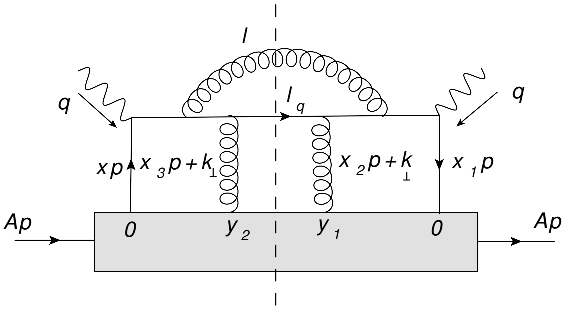


Figure 3.5: Example of HT contribution (rescattering of a quark with gluon radiation).

Combining such diagrams coherently, one can calculate the modification of the fragmentation function directly as a medium-dependent contribution:

$$D^{med}_{i\to h} = D^{vac}_{i\to h} + \Delta D^{med}_{i\to h} \tag{3.9}$$

The hierarchy of scales allows one to use the collinear approximation to factorize the fragmentation function and its modification from the hard scattering cross section. So, it may be generalized to the kinematics of a heavy-ion collision. This may be not true at very large jet energies where saturation effects become important.

The entire phenomenology of the medium is incorporated in the transport coefficient $\hat{q}$. The model used to fit the parameter to its space-time location is a Woods-Saxon distribution for cold matter or a 3-D hydrodynamical evolution for hot nuclear matter.

A disadvantage of this approach is the restriction to single scattering followed by single radiation in the medium, which makes this formalism more appropriate to thin media.

3.1.4 AMY

The Arnorld-Moore-Yaffe (AMY) approach describes parton energy loss in a hot equilibrated QGP, where the hierarchy $T \gg gT \gg g^2T$ is assumed. The hard on-shell parton, with energy several times that of the temperature, crossing such a medium, scatters off other partons in the medium with transverse momentum of the order of gT. Multiple scatterings of the incoming parton and the radiated gluon need to be combined to get the LO gluon radiation rate.

One essentially calculates the imaginary parts of ladder diagrams, like those shown in figure 3.6 by means of integral equations. The result is the $1 \to 2$ decay rates of a hard parton into a radiated gluon and another parton. These decay rates are evolved using a set of coupled Fokker-Planck like equations for quarks, antiquarks and gluons.

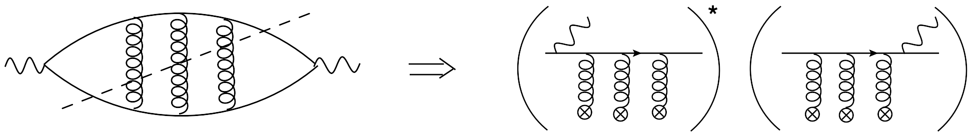


Figure 3.6: A typical ladder diagram in a thermal medium in the AMY formalism

This is the only approach that includes the absorption of thermal gluons as well as quark, antiquark pair annihilation and creation.

The medium modified FFs are obtained from the convolution of the vacuum FFs with the hard PDFs when exiting the plasma:

$$D^{med}_{a\to h}(z) = \int dp_f \frac{z'}{z} \sum_a (p_f, p_i) D^{vac}_{a\to h}(z') \tag{3.10}$$

where $z = p_h/p_i$ and $z' = p_h/p_f$, with p_i and p_f the momenta of the hard partons immediately after the hard scattering and before the exit from the plasma respectively.

Phenomena of vacuum radiation or vacuum-medium interference are not taken into account of this approach and its application to non-thermalised media is not well-established.

3.2 Jet Quenching

For the rest of this thesis, we will only focus on hard probes, with high-$p_\perp$, to describe the jet quenching phenomenon. Also, we will work in the massless limit and the framework used will be the BDMPS-Z/ASW. The first question we need to answer is how a highly energetic particle propagates through a dense medium. Therefore, in order to address to this issue, follows the most used formalism based on a semiclassical approach in which the medium can be considered as a background field since recoil is neglected. So, the interaction between the projectile partons and the target field is assumed to be eikonal, that is, the projectile partons propagate through the target without changing their transverse position but picking up an eikonal phase.

3.2.1 Eikonal Approximation

Lets consider an energetic hadronic projectile impinging on a large nuclear target. Choosing the Lorentz frame where the projectile moves in the positive $\hat{z}$ direction means that at high energy, the light-cone coordinate x^- of the projectile does not change during the propagation. So, the projectile can be characterized by a wave function whose relevant degrees of freedom are the transverse positions ($\mathbf{x_i}$) and the colour states of the partons:

$$\Psi_{in} = \sum_{\{\alpha_i \mathbf{x_i}\}} \Psi\left(\{\alpha_i,\ \mathbf{x_i}\}\right) |\{\alpha_i,\ \mathbf{x_i}\}\rangle \tag{3.11}$$

The colour index α_i can belong to the fundamental, anti-fundamental or adjoint representation of the colour group, corresponding to quark, antiquark or gluon in the wave function.

But the same effect happens during the propagation through the target and so, the projectile only sees the target fields at $x^- = 0$. This means we can consider the target field as independent of this component. Also, the propagation time is short, meaning that partons propagate independently of each other. For the same reason, the transverse positions of partons do not change during the propagation through the target. The only effect is the acquisition of an eikonal phase in the wave function due to the interaction with the gluonic field of the target. Thus, the projectile emerges from the interaction region with the wave function:

$$\Psi_{out} = \mathcal{S}\Psi_{in} = \sum_{\{\alpha_i \mathbf{x_i}\}} \Psi\left(\{\alpha_i \mathbf{x_i}\}\right) \prod_i W(\mathbf{x_i})_{\alpha_i \beta_i} \left|\{\beta_i,\ \mathbf{x_i}\}\right\rangle \tag{3.12}$$

where $\mathcal{S}$ is the S-matrix, and the W's are the Wilson lines [45–47] along the trajectories of the propagating particles trough a target gauge field $A^-(x_i^-, \mathbf{x_i})$:

$$W(\mathbf{x_i}) = \mathcal{P}\exp\left\{ig \int dx^+ T^a A_a^-(x^+, \mathbf{x_i})\right\} \tag{3.13}$$

$\mathcal{P}$ signals a path-ordering which tell us what is the order by which the particle cross the several layers of the target.

The interaction in the target field changes the relative phases between components of the wave function. Consequently the final state that emerges after the target is different from the initial one and can contain emitted gluons. Now, lets make a derivation of the Wilson line. For doing so, consider the diagram of one scattering, figure 3.7, where (p, i) and (p', k) are quark momenta and colour state before and after the scattering respectively.

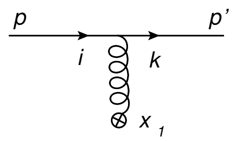


Figure 3.7: Scattering of a quark: one scattering centre

The amplitude is given by:

$$\mathcal{M} = \bar{u}(p') ig A_a^\mu(x^-, \mathbf{x}_\perp) T^a \gamma_\mu u(p) \tag{3.14}$$

The S-matrix is then:

$$\mathcal{S}_1(p, p') = \int d^4x e^{i(p'-p)\cdot x} u(p) ig A_a^\mu(x^-, \mathbf{x}_\perp) T^a \gamma_\mu u(p) \tag{3.15}$$

In the eikonal limit, because only a colour phase is exchanged with the medium, we can approximate in the norm $p' \simeq p$. If we average over initial spins and sum over final state spin we have:

$$\frac{1}{2}\sum_{\lambda}\bar{u}^{\lambda}(p)\gamma_{\mu}u^{\lambda}(p) = 2p_{\mu} \tag{3.16}$$

Also, because the transverse components are very small ($|\mathbf{p}_{\perp}| \ll p^{+}$) we can neglect them and write:

$$p_{\mu}A_{a}^{\mu} \simeq p^{+}A_{a}^{-} \tag{3.17}$$

With this, the gauge invariance is broken and so, one has to pay attention to the reference frame chosen.

Continuing:

$$\begin{aligned} \mathcal{S}_1(p,p') &\simeq ig\int dx_1^+dx_1^-d\mathbf{x_1}_{\perp}\mathrm{e}^{ix_1^+(p'-p)^-+ix_1^-(p'-p)^+-i\mathbf{x_1}_{\perp}\cdot(\mathbf{p}'-\mathbf{p})_{\perp}}2p^+A_a^-(x_1^+,\mathbf{x_1}_{\perp})T_{ki}^a \simeq \\ &\simeq (2\pi)\delta(p'-p)^+2p^+\int d\mathbf{x_1}_{\perp}\mathrm{e}^{-i\mathbf{x_1}_{\perp}\cdot(\mathbf{p}'-\mathbf{p})_{\perp}}\left[ig\int dx_1^+A_{ki}^-(x_1^-,\mathbf{x_1}_{\perp})\right] \end{aligned} \tag{3.18}$$

where we have used:

$$A_{ki}^-(x_1^-,\mathbf{x_1}_{\perp}) \equiv T_{ki}^aA_a^-(x_1^+,\mathbf{x_1}_{\perp}) \tag{3.19}$$

$$\int dx_1^-\mathrm{e}^{ix_1^-(p^+-p)^+} = (2\pi)\delta(p'-p)^+ \tag{3.20}$$

For two scatterings (fig. 3.8):

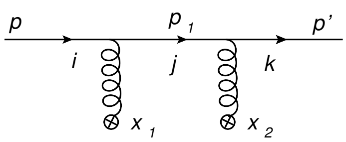


Figure 3.8: Scattering of a quark: two scattering centres

$$\mathcal{S}_2(p,p') = \int d^4x_1d^4x_2\frac{d^4p_1}{(2\pi)^4)}\mathrm{e}^{ix_1\cdot(p_1-p)}\mathrm{e}^{ix_2(p'-p)}\bar{u}(p')ig\gamma^{\nu}T_{kj}^{a_2}A_{\nu}^{a_2}(x_2)\frac{i\not{p}_1}{p_1^2+i\epsilon}ig\gamma^{\mu}T_{ji}^{a_1}A_{\mu}^{a_1}(x_1)u(p) \tag{3.21}$$

Using the eikonal limit we have $p \simeq p' \simeq p_1$ in the norm and $|\mathbf{p}_{\perp}| \sim 0$. Also, from Dirac equation we have:

$$\not{p}u(p) = \bar{u}(p')\not{p}' = 0 \tag{3.22}$$

So:

$$ip_1^{\alpha}\bar{u}(p)\gamma_{\nu}\gamma_{\alpha}\gamma_{\mu}u(p) = i2p_{\mu}2p_{\nu} \tag{3.23}$$

$$p_1^2 \simeq 2p_1^+ p_1^- \tag{3.24}$$

Using these results we will have for the $\mathcal{S}$-matrix:

$$\begin{aligned}
\mathcal{S}_2(p,p') = i(2\pi) \int dx_1^+ d\mathbf{x_1}_\perp dx_2^+ d\mathbf{x_2}_\perp \frac{dp_1^-}{2\pi} \frac{d\mathbf{p_1}}{(2\pi^2)} \frac{1}{2p^+ p_1^- + i\epsilon} \mathrm{e}^{ix_1^+(p_1-p)^- - i\mathbf{x_1}_\perp \cdot (\mathbf{p_1}-\mathbf{p})_\perp} \cdot \\
\cdot\, \mathrm{e}^{ix_2^+(p'-p_1)^- + ix_2^-(p'-p_1)^+ - i\mathbf{x_2}_\perp \cdot (\mathbf{p}'-\mathbf{p_1})_\perp} ig2p^+ A_{kj}^-(x_2^+, \mathbf{x_2}_\perp) ig2p^+ A_{ji}^-(x_1^+, \mathbf{x_1}_\perp) = \\
= (2\pi)\delta(p'-p)^+ 2p^+ \int dx_1^+ d\mathbf{x_1}_\perp dx_2^+ d\mathbf{x_2}_\perp \frac{dp_1^-}{2\pi} \frac{d\mathbf{p_1}_\perp}{(2\pi)^2} \frac{i}{p_1^- + i\epsilon} \mathrm{e}^{ip_1^-(x_1-x_2)^+ - \mathbf{p_1}_\perp \cdot (\mathbf{x_1}-\mathbf{x_2})} \cdot \\
\cdot\, \mathrm{e}^{-ix_1^+ p^- + i\mathbf{x_1}_\perp \cdot \mathbf{p}_\perp} \mathrm{e}^{ix_2^+ p'^- - i\mathbf{x_2}_\perp \cdot \mathbf{p}'_\perp} igA_{ji}^-(x_1^+, \mathbf{x_1}_\perp) igA_{kj}^-(x_2^+, \mathbf{x_2}_\perp)
\end{aligned} \tag{3.25}$$

For the integral in p_1^- we will have to use the residues theorem. In order to get the contribution from the pole, we have to have $x_1 < x_2$ like shown in figure 3.9.

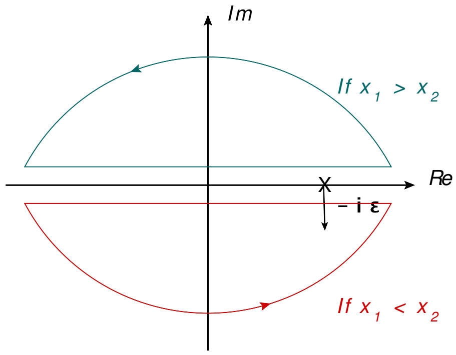


Figure 3.9: Contours of the poles

This means we need to introduce the Heaviside step functions defined as:

$$\theta(x) = \begin{cases} 0 & ,x < 0 \\ 1 & ,x \geq 0 \end{cases} \tag{3.26}$$

The result of the integral should be:

$$\int \frac{dp_1^-}{2\pi} \frac{i}{p_1^- + i\epsilon} \mathrm{e}^{ip_1^-(x_1-x_2)^+} = \theta(x_2 - x_1)^+ \tag{3.27}$$

With these, our expression takes the form:

$$\begin{aligned}
\mathcal{S}_2(p,p') =& (2\pi)\delta(p'-p)^+ 2p^+ \int dx_1^+ dx_2^+ d\mathbf{x_2}_\perp \theta(x_2-x_1)^+ \mathrm{e}^{-ix_1^+ p^- + ix_2^+ p'^-} \mathrm{e}^{i\mathbf{x_2}_\perp \cdot (\mathbf{p}-\mathbf{p}')_\perp} \cdot \\
& \cdot igA_{ji}^-(x_1^+, \mathbf{x_2}_\perp) igA_{kj}^-(x_2^+, \mathbf{x_2}_\perp)
\end{aligned} \tag{3.28}$$

Using the general formula:

$$\int dx_1^+ dx_2^+ \dots dx_n^+ A^-(x_1^+, \mathbf{x_1}_\perp)\theta(x_2^+ - x_1^+)A^-(x_2^+, \mathbf{x_2}_\perp)\dots\theta(x_n^+ - x_{n-1}^+)A^-(x_n^+, \mathbf{x_n}_\perp) = \\ = \frac{1}{n!}\mathcal{P}\left[\int dx^+ A^-(x^+, \mathbf{x}_\perp)\right]^n \tag{3.29}$$

we can get easily the final result:

$$\mathcal{S}_2(p,p') = (2\pi)\delta(p'-p)^+ 2p^+ \int \mathbf{x}_\perp \mathrm{e}^{-i\mathbf{x}_\perp\cdot(\mathbf{p}'-\mathbf{p})_\perp} \frac{1}{2}\mathcal{P}\left[ig\int dx^+ A^-_{ki}(x^+, \mathbf{x}_\perp)\right]^2 \tag{3.30}$$

The generalization to n scatterings is straightforward:

$$\mathcal{S}_n(p,p') = (2\pi)\delta(p'-p)^+ 2p^+ \int d\mathbf{x}_\perp \mathrm{e}^{-i\mathbf{x}_\perp\cdot(\mathbf{p}'-\mathbf{p}_\perp)} \frac{1}{n!}\mathcal{P}\left[ig\int dx^+ A^-_{ki}(x^+, \mathbf{x}_\perp)\right]^n \tag{3.31}$$

In order to get the total $\mathcal{S}$-matrix for the multiple scattering process we have to sum over all scattering centres. Doing so, the Wilson line (3.13) appears automatically:

$$\mathcal{S}(p,p') = \sum_{n=0}^{\infty}\mathcal{S}_n(p,p') = (2\pi)\delta(p'-p)^+ 2p^+ \int d\mathbf{x}_\perp \mathrm{e}^{-i\mathbf{x}_\perp\cdot(\mathbf{p}'-\mathbf{p}_\perp)}\mathcal{P}\exp\left[ig\int_{x_0^+}^{L^+} dx^+ A^-_{ki}(x^+, \mathbf{x}_\perp)\right] = \\ = (2\pi)\delta(p'-p)^+ 2p^+ \int d\mathbf{x}_\perp \mathrm{e}^{-i\mathbf{x}_\perp\cdot(\mathbf{p}'-\mathbf{p}_\perp)} W_{ki}(\mathbf{x}_\perp; x_0^+, L^+) \tag{3.32}$$

3.2.2 First Correction to eikonal approximation

However, there are situations in which the restrictions of the previous formulation need to be relaxed to allow some movement in the transverse plane of the propagating particle. One situation where this can be applied is the medium-induced gluon radiation where the gluon position follows a Brownian motion in the transverse plane.

To derive the new changes to the Wilson line, it is just to keep terms with $p_\perp^2$ in the norm. In this way, we will find an expression for two scatterings very similar to equation (3.25):

$$\mathcal{S}_2(p,p') = \int dx_1^+ d\mathbf{x_1}_\perp dx_2^+ dx_2^- d\mathbf{x_2}_\perp \frac{dp_1^-}{2\pi}\frac{d\mathbf{p_1}_\perp}{(2\pi)^2}\frac{i}{2p^+p_1^- - p_\perp^2 + i\epsilon}\mathrm{e}^{ix_1^+(p_1-p)^- - i\mathbf{x_1}_\perp\cdot(\mathbf{p_1}-\mathbf{p})}\cdot \\ \cdot \mathrm{e}^{ix_2^+(p'-p_1)^- + ix_2^-(p'-p)^+ - i\mathbf{x_2}_\perp\cdot(\mathbf{p}'-\mathbf{p})_\perp} ig2p^+ A^-_{kj}(x_2^+, \mathbf{x_2}_\perp) ig2p^+ A^-_{ji}(x_1^+, \mathbf{x_1}_\perp) \\ = (2\pi)\delta(p'-p)^+ 2p^+ \int dx_1^+ d\mathbf{x_1}_\perp dx_2^+ d\mathbf{x_2}_\perp \frac{dp_1^-}{2\pi}\frac{d\mathbf{p_1}_\perp}{(2\pi)^2}\frac{i}{p_1^- - (p_{1\perp}^2/2p^+ - i\epsilon)}\cdot \\ \cdot \mathrm{e}^{ip_1^-(x_1-x_2)^+ - i\mathbf{p_1}_\perp\cdot(\mathbf{x_1}-\mathbf{x_2})_\perp}\mathrm{e}^{ix_1^+p^- + i\mathbf{x_1}_\perp\cdot\mathbf{p}_\perp}\mathrm{e}^{ix_2^+p'^- - i\mathbf{x_2}_\perp\cdot\mathbf{p}'_\perp}\cdot \\ \cdot igA^-_{kj}(x_2^+, \mathbf{x_2}_\perp) igA^-_{ji}(x_1^+, \mathbf{x_1}_\perp) \tag{3.33}$$

Using the residues theorem we will get:

$$\int \frac{dp_1^-}{2\pi} \frac{i}{p_1^- - (p_{1_\perp}^2/2p^+ - i\epsilon)} \mathrm{e}^{ip_1^-(x_1-x_2)^+} = \theta(x_2 - x_1)\mathrm{e}^{i\frac{p_{1_\perp}^2}{2p^+}(x_1-x_2)^2} \tag{3.34}$$

For the integration in $\mathbf{p_1}_\perp$, we have to complete the square:

$$\begin{aligned}
&\int \frac{d\mathbf{p_1}_\perp}{(2\pi)^2} \exp\left[i\frac{p_{1_\perp}^2}{2p^+}(x_1-x_2)^+ - i\mathbf{p_1}_\perp \cdot (\mathbf{x_1} - \mathbf{x_2})_\perp\right] = \\
&= \int \frac{\mathbf{p_1}_\perp}{(2\pi)^2} \exp\left\{\frac{i(x_1-x_2)^+}{2p^+}\left[p_{1_\perp}^2 - i\mathbf{p_1}_\perp \cdot \frac{(\mathbf{x_1}-\mathbf{x_2})_\perp}{(x_1^+ - x_2^+)}2p^+\right]\right\} = \\
&= \int \frac{d\mathbf{p_1}_\perp}{(2\pi)^2} \exp\left[\frac{-i(x_2-x_1)^+}{2p^+}p_{1_\perp}^2\right] \exp\left[\frac{-ip^+}{2}\frac{(\mathbf{x_1}-\mathbf{x_2})_\perp^2}{(x_1^+ - x_2^+)}\right]
\end{aligned} \tag{3.35}$$

Now, using the gaussian integral in two dimensions [48] which tell us that:

$$\int \exp\left[-\frac{1}{2}A_{ij}x^i x^j\right] d^2x = \sqrt{\frac{(2\pi)^2}{\det(A)}} \tag{3.36}$$

we find the path-integral formula [46, 47]:

$$\begin{aligned}
&\int \frac{d\mathbf{p_1}_\perp}{(2\pi)^2} \exp\left[i\frac{p_{1_\perp}^2}{2p^+}(x_1^+ - x_2^+).i\mathbf{p_1}_\perp \cdot (\mathbf{x_1}-\mathbf{x_2})_\perp\right] = \frac{p^+}{2\pi i(x_2^+ - x_1^+)} \exp\left[\frac{ip^+}{2}\frac{(\mathbf{x_2}-\mathbf{x_1})_\perp^2}{(x_2^+ - x_1^+)}\right] \equiv \\
&\equiv \int_{\mathbf{r}(x_1^+)=\mathbf{x_1}_\perp}^{\mathbf{r}(x_2^+)=\mathbf{x_2}_\perp} \mathcal{D}\mathbf{r}(\xi) \exp\left[\frac{ip^+}{2}\int_{x_1^+}^{x_2^+} d\xi \dot{\mathbf{r}}^2(\xi)\right] \equiv G_0(x_1^+, \mathbf{x_1}_\perp; x_2^+, \mathbf{x_2}_\perp)
\end{aligned} \tag{3.37}$$

which can be associated with the Feynman propagator of a free particle that propagates in the transverse plane from $\mathbf{x_1}_\perp$ at time x_1^+ to $\mathbf{x_2}_\perp$ at time x_2^+. There, the parameter $\dot{\mathbf{r}} = d\mathbf{r}/dx^+$ and the boundary conditions on the free path integral are $\mathbf{r}(x_i^+) = \mathbf{x_i}$. For path integral formalism, see, for example, [7, 49].

Our $\mathcal{S}$-matrix takes the form:

$$\begin{aligned}
\mathcal{S}_2(p,p') =& (2\pi)\delta(p'-p)^+ 2p^+ \int dx_1^+ d\mathbf{x_1}_\perp dx_2^+ d\mathbf{x_2}_\perp \theta(x_2-x_1)^+ G_0(x_1^+, \mathbf{x_1}_\perp; x_2^+, \mathbf{x_2}_\perp) \cdot \\
&\cdot \mathrm{e}^{i\mathbf{x_1}_\perp \cdot \mathbf{p}_\perp - i\mathbf{x_2}_\perp \cdot \mathbf{p'}_\perp} igA_{kj}^-(x_2^+, \mathbf{x_2}_\perp) igA_{ji}^-(x_1^+, \mathbf{x_1}_\perp) \mathrm{e}^{ix_2^+ p'^-}
\end{aligned} \tag{3.38}$$

The total amplitude for this process is then:

$$
\begin{aligned}
S(p,p') = (2\pi)\delta(p'-p)^{+}2p^{+}\sum_{n=0}^{\infty}\Big\{\int dx_1^{+}d\mathbf{x_1}_{\perp}\dots dx_n^{+}d\mathbf{x_n}_{\perp}\, igA^{-}(x_1^{+},\mathbf{x_1}_{\perp})\theta(x_2-x_1)^{+}\cdot \\
\cdot\, G_0(x_1^{+},\mathbf{x_1}_{\perp};x_2^{+},\mathbf{x_2}_{\perp}|p^{+})igA^{-}(x_2,\mathbf{x_2}_{\perp})\dots igA^{-}(x_{n-1}^{+},\mathbf{x_{(n-1)}}_{\perp})\theta(x_n-x_{n-1})^{+}\cdot \\
\cdot\, G_0(x_{n-1}^{+},\mathbf{x_{(n-1)}}_{\perp};x_n^{+},\mathbf{x_n}_{\perp}|p^{+})igA^{-}(x_n^{+},\mathbf{x_n}_{\perp})\Big\}\, \mathrm{e}^{i\mathbf{x_1}_{\perp}\cdot\mathbf{p}_{\perp}-i\mathbf{x_n}_{\perp}\cdot\mathbf{p}'_{\perp}}\mathrm{e}^{ix_n^{+}p'^{-}} \simeq \\
\simeq (2\pi)\delta(p'-p)^{+}2p^{+}\int d\mathbf{x_1}_{\perp}d\mathbf{x_n}_{\perp}\, G(x_1^{+},\mathbf{x_1}_{\perp};x_n^{+},\mathbf{x_n}_{\perp})\mathrm{e}^{-i\mathbf{x_1}_{\perp}\cdot\mathbf{p}_{\perp}}\mathrm{e}^{ix_n^{+}p'^{-}-i\mathbf{x_n}_{\perp}\cdot\mathbf{p}'_{\perp}}
\end{aligned}
\tag{3.39}
$$

where the full Green's function is [46, 47]:

$$
\begin{aligned}
G(x_1^{+},\mathbf{x_1}_{\perp};x_n^{+},\mathbf{x_n}_{\perp}|p^{+}) &= \int \mathcal{D}\mathbf{r}\exp\left\{\frac{ip^{+}}{2}\int d\xi\left[\frac{d\mathbf{r}}{d\xi}\right]^{2}\right\}\cdot\mathcal{P}\exp\left\{i\int_{x_1^{+}}^{x_n^{+}}d\xi A^{-}(\xi,\mathbf{r}(\xi))\right\} = \\
&= \int \mathcal{D}\mathbf{r}\exp\left\{\frac{ip^{+}}{2}\int d\xi\dot{\mathbf{r}}^{2}(\xi)\right\}W(\mathbf{r}(\xi);x_1^{+},x_n^{+})
\end{aligned}
\tag{3.40}
$$

This path integral describes the Brownian motion of a particle in the transverse plane, where the free transverse motion is governed by the kinematic term $p^{+}\dot{\mathbf{r}}^{2}/2$. This is a two-dimensional light-cone hamiltonian with *mass* given by the total energy p^{+} of the particle along the beam. During the motion, the wave function of the particle also acquires a phase given by the Wilson line along the trajectory. Therefore, the particle follows a non-eikonal trajectory inside the medium. It should be noted that this is the only correction to be made if the medium is considered to be static.

The infinite energy limit corresponds to the classical limit, where a single straight classical trajectory dominates the sum over paths. In this limit, the Green's function (3.40) reduces to the eikonal Wilson line in (3.13):

$$
\lim_{p^{+}\to\infty} G(x_1^{+},\mathbf{x_1}_{\perp};x_n^{+},\mathbf{x_n}_{\perp}|p^{+}) = \delta^{2}(\mathbf{x_1}-\mathbf{x_n})_{\perp}\theta(x_1^{+}-x_n^{+})W(\mathbf{x_1}_{\perp};x_1^{+},x_n^{+}) \tag{3.41}
$$

3.2.3 Medium Averages

Since the time of the propagation of an energetic projectile through the target is short, we can say that the gluonic field in the target does not vary during the time of the interaction. This is true only if the time scale on which the target fields vary is greater than the propagation time, which we will assume to be true. With this assumption, we can say that each scatterings happens on a frozen field profile. But in order to calculate the cross section, one has to average over the field ensemble which represents the wave function of the target.

There are various averaging procedures discussed in the literature, but we will follow the one from [45–47]. Here, the target is pictured as an assembly of static coloured field sources randomly distributed within the target longitudinal and transverse extent. Scattering centres are independent of each other and thus completely decorrelated so that no colour flow appears between sources separated by more than a distance $1/\mu$, μ being a typical scale in the medium as the Debye screening length. We are then assuming that this distance is much shorter than the mean free path of particles through the target.

We can define the scattering potential of each scattering centre at positions $(x_n^-, \mathbf{x_n}_\perp)$ to be $a^{-a}(\mathbf{q})$. In the high energy approximation, each scattering centre transfers only transverse momentum to the projectile:

$$A^{-a}(x^+, \mathbf{x}_\perp) = \sum_n \int \frac{d\mathbf{q}_\perp}{(2\pi)^2} \mathrm{e}^{i(\mathbf{x}-\mathbf{x_n})_\perp \cdot \mathbf{q}_\perp} a^{-a}(\mathbf{q}) \delta(x - x_n)^+ \tag{3.42}$$

and so, a projectile can only interact with sources localized in $x^+ = x_n^+$ like shown in figure 3.10:

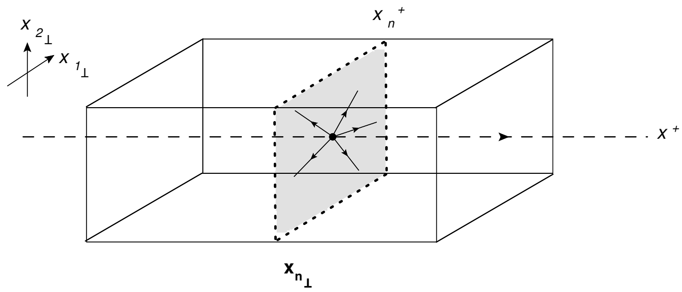


Figure 3.10: Representation of particle interaction with the medium.

The medium averages are done by integrating in the longitudinal x_n^+ and transverse coordinates $\mathbf{x_{n_\perp}}$ of the scattering centres:

$$\begin{aligned} \left\langle A^{-a}(x^, \mathbf{x}_\perp) A^{-b}(y^+, \mathbf{y}_\perp) \right\rangle = \sum_n \int \frac{d\mathbf{q}_\perp}{(2\pi)^2} \frac{d\mathbf{k}_\perp}{(2\pi)^2} dx^+ d\mathbf{x_{n_\perp}} \mathrm{e}^{i(\mathbf{x}-\mathbf{x_n})_\perp \cdot \mathbf{q}_\perp + i(\mathbf{y}-\mathbf{x_n})_\perp \cdot \mathbf{k}}. \\ \cdot\, a^{-a}(\mathbf{q}) a^{-b}(\mathbf{k}) \delta(x - x_n)^+ \delta(y - x_n)^+ \end{aligned} \tag{3.43}$$

Introducing the longitudinal density of scattering centres $n(x^+) = \int dx_n^+ \delta(x^+ - x_n^+)$:

$$\begin{aligned} \left\langle A^{-a}(x^, \mathbf{x}_\perp) A^{-b}(y^+, \mathbf{y}_\perp) \right\rangle = \left\langle A^{-a}(x^, \mathbf{x}_\perp) \left(A^{-b}\right)^* (y^+, \mathbf{y}_\perp) \right\rangle = \\ = n(x^+) \delta(x - y)^+ \delta^{ab} \frac{1}{(2\pi)^2} \int d\mathbf{q} \mathrm{e}^{i(\mathbf{x}-\mathbf{y})_\perp \cdot \mathbf{q}_\perp} \left| a(\mathbf{q}) \right|^2 \end{aligned} \tag{3.44}$$

For the cross section, we are interested in quantities like:

$$\left\langle \mathrm{Tr} \left[W^\dagger(\mathbf{x}_\perp) W(\mathbf{y}_\perp) \right] \right\rangle_t \equiv \mathrm{Tr} \left\langle W^\dagger(\mathbf{x}_\perp) W(\mathbf{y}_\perp) \right\rangle \tag{3.45}$$

with Wilson lines in the fundamental or adjoint representation. Expanding the exponents and taking the contribution until the quadratic terms in g, we have for the fundamental representation:

$$
\begin{aligned}
\mathrm{Tr}\left\langle W(\mathbf{x}_\perp)W^\dagger(\mathbf{y}_\perp)\right\rangle = \Big\langle\Big(&1 - ig\int dy^+ \left(A^{-b}\right)^* (y^+,\mathbf{y}_\perp)T^b + ig\int dx^+ A^{-a}(x^+,\mathbf{x}_\perp)T^a - \\
&- \frac{g^2}{2}\int dy_1^+ dy_2^+ \left(A^{-b_1}\right)^* (y_1^+,\mathbf{y}_\perp)\left(A^{-b_2}\right)^+ (y_2^+,\mathbf{y}_\perp)T^{b_1}T^{b_2} - \\
&- \frac{g^2}{2}\int dx_1^+ dx_2^+ A^{-a_1}(x_1^+,\mathbf{x}_\perp)A^{-a_2}(x_2^+,\mathbf{x}_\perp)T^{a_1}T^{a_2} + \\
&+ g^2\int dx^+ dy^+ A^{-a}(x^+,\mathbf{x}_\perp)\left(A^{-b}\right)^+ (y^+,\mathbf{y}_\perp)\mathrm{Tr}\left[T^aT^b\right]\Big)\Big\rangle = \\
= N_c\Big\{1 - \frac{1}{2}&\int d\xi^+ n(\xi^+)\frac{1}{(2\pi)^2}\int d\mathbf{q}\left[1 - \mathrm{e}^{i(\mathbf{x}-\mathbf{y})_\perp\cdot\mathbf{q}_\perp}\right]C_F\Big\}
\end{aligned}
\tag{3.46}
$$

Introducing the dipole cross section as:

$$
\sigma(\mathbf{x}_\perp - \mathbf{y}_\perp) = \frac{g^2}{2\pi^2}\int d\mathbf{q}\,|a(\mathbf{q})|^2\left[1 - \mathrm{e}^{i(\mathbf{x}-\mathbf{y})_\perp\cdot\mathbf{q}}\right] \tag{3.47}
$$

our result will be:

$$
\frac{1}{N_c}\mathrm{Tr}\left\langle W(\mathbf{x}_\perp)W^\dagger(\mathbf{y}_\perp)\right\rangle = 1 - \frac{C_F}{2}\int d\xi^+ n(\xi^+)\sigma(\mathbf{x}_\perp - \mathbf{y}_\perp) \tag{3.48}
$$

If we had expanded equation (3.46) in all orders, our result, for both representations, would be:

$$
\frac{1}{N_c}\mathrm{Tr}\left\langle W_F(\mathbf{x}_\perp)W_F^\dagger(\mathbf{y}_\perp)\right\rangle = \exp\left\{-\frac{C_F}{2}\int d\xi^+ n(\xi^+)\sigma(\mathbf{x}_\perp - \mathbf{y}_\perp)\right\} \tag{3.49}
$$

$$
\frac{1}{N_c^2-1}\mathrm{Tr}\left\langle W_A(\mathbf{x}_\perp)W_A^\dagger(\mathbf{y}_\perp)\right\rangle = \exp\left\{-\frac{C_A}{2}\int d\xi^+ n(\xi^+)\sigma(\mathbf{x}_\perp - \mathbf{y}_\perp)\right\} \tag{3.50}
$$

A possible generalization of these results is:

$$
\left\langle W_F^{AB}(\mathbf{x}_\perp)W_F^{\dagger BC}(\mathbf{y}_\perp)\right\rangle = \frac{\delta^{AC}}{N_c}\mathrm{Tr}\left\langle W_F(\mathbf{x}_\perp)W_F^\dagger(\mathbf{y}_\perp)\right\rangle \tag{3.51}
$$

$$
\left\langle W_A^{ab}(\mathbf{x}_\perp)W_A^{\dagger bc}(\mathbf{y}_\perp)\right\rangle = \frac{\delta^{ac}}{N_c^2-1}\mathrm{Tr}\left\langle W_A(\mathbf{x}_\perp)W_A^\dagger(\mathbf{y}_\perp)\right\rangle \tag{3.52}
$$

But these are not the only terms we will find in the radiated spectrum. Because we will use the correction to the eikonal approximation (3.40), another averages over the medium sources will appear. Lets derive some analytical expressions for some useful expressions, where the quadratic Casimir C_F or C_A will be implicit in the definition of the dipole cross section:

$$A = \frac{1}{N_c^2-1}\mathrm{Tr}\left\langle G(x^+,\mathbf{x}_\perp;y^+,\mathbf{y}_\perp|\omega)W^\dagger(\mathbf{z}_\perp;x^+,y^+)\right\rangle = \int_{\mathbf{r}_\perp(x^+)=\mathbf{x}_\perp}^{\mathbf{r}_\perp(y^+)=\mathbf{y}_\perp}\mathcal{D}\mathbf{r}_\perp(\xi)\cdot$$
$$\cdot\frac{1}{N_c^2-1}\mathrm{Tr}\left\langle W(\mathbf{r}_\perp;x^+,y^+)W^\dagger(\mathbf{z}_\perp;x^+,y^+)\right\rangle =$$
$$= \int_{\mathbf{r}_\perp(x^+)=\mathbf{x}_\perp}^{\mathbf{r}_\perp(y^+)=\mathbf{y}_\perp}\mathcal{D}\mathbf{r}_\perp(\xi)\exp\left\{\int_{x^+}^{y^+}d\xi\left[\frac{i\omega}{2}\dot{\mathbf{r}}_\perp^2(\xi)-\frac{1}{2}n(\xi)\sigma\left(\mathbf{r}_\perp[\xi]-\mathbf{z}_\perp\right)\right]\right\} \tag{3.53}$$

Making the following change of variables,

$$\mathbf{r}_\perp(\xi)-\mathbf{z}_\perp = \mathbf{v}_\perp(\xi) \Leftrightarrow \mathbf{r}_\perp(\xi) = \mathbf{v}_\perp(\xi)+\mathbf{z}_\perp \Rightarrow \mathcal{D}\mathbf{r}_\perp(\xi) = \mathcal{D}\mathbf{v}_\perp(\xi) \tag{3.54a}$$
$$\dot{\mathbf{r}}_\perp^2(\xi) = \left(\dot{\mathbf{v}}_\perp(\xi)+\dot{\mathbf{z}}_\perp\right)^2 = \dot{\mathbf{v}}_\perp^2(\xi) \tag{3.54b}$$
$$\mathbf{v}_\perp(x^+) = \mathbf{r}_\perp(x^+)-\mathbf{z}_\perp = \mathbf{x}_\perp-\mathbf{z}_\perp \text{ and } \mathbf{v}_\perp(y^+) = \mathbf{r}_\perp(y^+)-\mathbf{z}_\perp = \mathbf{y}_\perp-\mathbf{z}_\perp \tag{3.54c}$$

and defining,

$$\mathcal{K}_\omega(x^+,\mathbf{x}_\perp;y^+,\mathbf{y}_\perp) = \int\mathcal{D}\mathbf{r}_\perp\exp\left[\int_{x^+}^{y^+}d\xi\left(\frac{i\omega}{2}\dot{\mathbf{r}}_\perp^2(\xi)-\frac{1}{2}n(\xi)\sigma(\mathbf{r}_\perp)\right)\right] \tag{3.55}$$

we will get:

$$A = \mathcal{K}_\omega(x^+,\mathbf{x}_\perp-\mathbf{z}_\perp;y^+,\mathbf{y}_\perp-\mathbf{z}_\perp) \tag{3.56}$$

Another possibility is:

$$B = \frac{1}{N_c^2-1}\mathrm{Tr}\left\langle G^\dagger(x^+,\mathbf{x}_\perp;y^+,\mathbf{y}_\perp|\omega)G(x^+,\mathbf{z}_\perp;y^+,\mathbf{w}_\perp|\omega)\right\rangle = \int_{\mathbf{r}_\perp(x^+)=\mathbf{x}_\perp}^{\mathbf{r}_\perp(y^+)=\mathbf{y}_\perp}\mathcal{D}\mathbf{r}_\perp(\xi)\int_{\mathbf{u}_\perp(x^+)=\mathbf{z}_\perp}^{\mathbf{u}_\perp(y^+)=\mathbf{w}_\perp}\mathcal{D}\mathbf{u}_\perp(\xi)\cdot$$
$$\cdot\exp\left\{\frac{i\omega}{2}\int_{x^+}^{y^+}d\xi\left[\dot{\mathbf{u}}_\perp^2(\xi)-\dot{\mathbf{r}}_\perp^2(\xi)\right]\right\}\frac{1}{N_c^2-1}\mathrm{Tr}\left\langle W^\dagger(\mathbf{r}_\perp(\xi);x^+,y^+)W(\mathbf{u}_\perp(\xi);x^+,y^+)\right\rangle =$$
$$= \int\mathcal{D}\mathbf{r}_\perp(\xi)\int\mathcal{D}\mathbf{u}_\perp(\xi)\exp\left\{\int_{x^+}^{y^+}d\xi\left[\frac{i\omega}{2}\left(\dot{\mathbf{u}}_\perp^2[\xi]-\dot{\mathbf{r}}_\perp^2[\xi]\right)-\frac{1}{2}n(\xi)\sigma\left(\mathbf{u}_\perp[\xi]-\mathbf{r}_\perp[\xi]\right)\right]\right\} \tag{3.57}$$

Making the change of variables:

$$\left\{\begin{array}{l}\boldsymbol{\alpha}_\perp(\xi) = \mathbf{u}_\perp(\xi)+\mathbf{r}_\perp(\xi)\\ \boldsymbol{\beta}_\perp(\xi) = \mathbf{u}_\perp(\xi)-\mathbf{r}_\perp(\xi)\end{array}\right. \Leftrightarrow \left\{\begin{array}{l}\mathbf{r}_\perp(\xi) = \frac{\boldsymbol{\alpha}_\perp-\boldsymbol{\beta}_\perp}{2}\\ \mathbf{u}_\perp(\xi) = \frac{\boldsymbol{\alpha}_\perp+\boldsymbol{\beta}_\perp}{2}\end{array}\right. \tag{3.58a}$$
$$\Rightarrow \mathcal{D}\mathbf{r}_\perp(\xi)\mathcal{D}\mathbf{r}_\perp(\xi) = \mathcal{D}\boldsymbol{\alpha}_\perp(\xi)\mathcal{D}\boldsymbol{\beta}\perp(\xi) \tag{3.58b}$$

$$\begin{cases} \boldsymbol{\alpha}_\perp(x^+) = \mathbf{u}_\perp(x^+) + \mathbf{r}_\perp(x^+) = \mathbf{z}_\perp + \mathbf{x}_\perp \\ \boldsymbol{\alpha}_\perp(y^+) = \mathbf{u}_\perp(y^+) + \mathbf{r}_\perp(y^+) = \mathbf{w}_\perp + \mathbf{y}_\perp \\ \boldsymbol{\beta}_\perp(x^+) = \mathbf{u}_\perp(x^+) - \mathbf{r}_\perp(x^+) = \mathbf{z}_\perp - \mathbf{x}_\perp \\ \boldsymbol{\beta}_\perp(y^+) = \mathbf{u}_\perp(y^+) - \mathbf{r}_\perp(y^+) = \mathbf{w}_\perp - \mathbf{y}_\perp \end{cases} \tag{3.59a}$$

$$\Rightarrow \dot{\mathbf{u}}_\perp^2(\xi) - \dot{\mathbf{r}}_\perp^2(\xi) = \dot{\boldsymbol{\alpha}}_\perp(\xi)\dot{\boldsymbol{\beta}}_\perp(\xi) \tag{3.59b}$$

we will end with the following result [50]

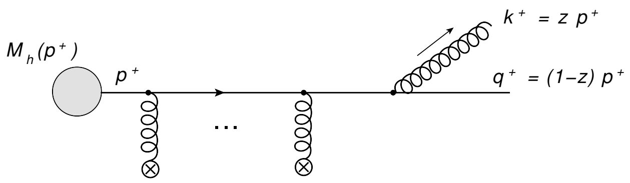
$$B = \left(\frac{\omega}{2\pi(x-y)^+}\right)^2 \exp\left\{\frac{i\omega}{2(x-y)^+}\left[(\mathbf{x}-\mathbf{y})_\perp^2 - (\mathbf{z}-\mathbf{w})_\perp^2\right] - \frac{1}{2}\int_{x^+}^{y^+} d\xi n(\xi)\sigma\left(\boldsymbol{\mathcal{L}}_\perp[\xi]\right)\right\} \tag{3.60}$$

where,

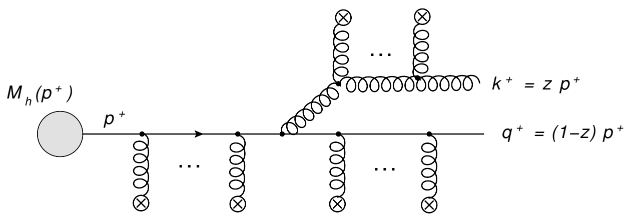
$$\mathcal{L}_\perp = \frac{1}{(x-y)^+}\left[(\xi - y^+)\boldsymbol{\alpha}_\perp(x^+) + (x^+ - \xi)\boldsymbol{\alpha}_\perp(y^+)\right] \tag{3.61}$$

3.2.4 Spectrum of Radiated Gluons in the Presence of a Medium

Now that we have all ingredients, we may follow for the main purpose of this thesis: to compute the medium modifications to the vacuum splitting functions of chapter 2, equations (2.61). So, we need to determine the amplitude for the process of medium-induced gluon radiation. There will be two diagrams: the one in which the gluon is radiated outside the medium (fig. 3.11(a)) since it will be affected by colour rotations suffered by the quark; and the one in which the gluon is radiated inside the medium (fig. 3.11(b)) and also suffering multiple soft scattering:



(a)



(b)

Figure 3.11: Medium-induced gluon radiation diagrams.

For these jet quenching studies, we will assume that a fast particle is produced at a given point inside the medium where the hard process, h, takes place. We will work in the approximation $p^+ \gg k^+ \gg |\mathbf{k}_\perp|^2$, i.e., $z \sim 0$, but keeping terms in $k_\perp^2/k^+$ as explained in section 3.2.2.

We will start by the two most trivial processes: the elastic process and the inelastic process in the vacuum. First of all, we will write the amplitude for the elastic process (fig. 3.12) which is:

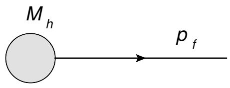


Figure 3.12: Diagram of the elastic process.

$$\mathcal{S}_{el}(p_i, p_f) = \bar{u}(p_f) M_h(p_i) \tag{3.62}$$

where $M_h(p_i)$ has a structure of a spinor $u(p_i)$.

The inelastic process in the vacuum should be (fig. 3.13):

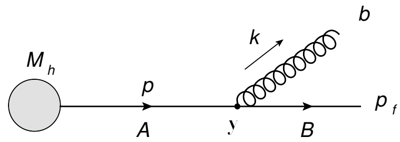


Figure 3.13: Gluon radiation in the vacuum.

$$\mathcal{S}_{rad}^{(0)}(p_f, k) = \int d^4x_0 d^4y \frac{d^4p}{(2\pi)^4} \bar{u}(p_f) ig\gamma^{\mu} T_{BA}^b \epsilon_{\mu}^{*b}(k) \frac{i\not{p}}{p^2 + i\epsilon} M_h(x_0) \mathrm{e}^{iy\cdot(p_f+k-o)} \mathrm{e}^{ix_0\cdot p} \tag{3.63}$$

We can pass from the space of coordinates to the space of momenta through the transformation,

$$\int d^4x_0 \mathrm{e}^{ix_0\cdot p} M_h(x_0) = M_h(p) \tag{3.64}$$

meaning that,

$$\mathcal{S}_{rad}^{(0)}(p_f, k) = \bar{u}(p_f) if\gamma^{\mu} T_{BA}^b \epsilon_{\mu}^{*b} \frac{i(\not{p}_f + \not{k})}{(p_f + k)^2 + i\epsilon} M_h(p) \tag{3.65}$$

Neglecting terms proportional to k, the Dirac structure can be simplified to:

$$\bar{u}(p_f)\not{\epsilon}^{*b}(\not{p}_f + \not{k}) \simeq 2\epsilon^{*b}(k) \cdot p_f \bar{u}(p_f) \tag{3.66}$$

$$\frac{2\epsilon^{*b}(k) \cdot p_f}{(p_f + k)^2} = \frac{2\epsilon^{+b}(k) \cdot p_f}{2p_f \cdot k} = \frac{2\boldsymbol{\epsilon}_{\perp}^{*b}(k) \cdot \mathbf{k}_{\perp}}{k_{\perp}^2} \tag{3.67}$$

The final result will be:

$$\mathcal{S}_{rad}^{(0)}(p_f, k) = -2gT_{BA}^b \frac{\boldsymbol{\epsilon}_{\perp}^{*b} \cdot \mathbf{k}_{\perp}}{k_{\perp}^2} \mathcal{S}_{el}(p_f + k, p_f) \tag{3.68}$$

Because we only want information from the radiated part, we must divide our result by the elastic channel:

$$\frac{S_{rad}^{(0)}}{S_{el}} = M_{vac} \simeq -2gT_{BA}^{b}\frac{\boldsymbol{\epsilon}_{\perp}^{*b}\cdot \mathbf{k}_{\perp}}{k_{\perp}^{2}} \tag{3.69}$$

This means that from the cross section equation (2.22), the only phase factor that remains is:

$$\frac{d^3k}{2k^0(2\pi)^3} \equiv \frac{dk^+d^2k_\perp}{2k^+(2\pi)^3} \tag{3.70}$$

With this, we can find the double differential spectrum of radiated gluons:

$$\overline{|M_{vac}|^2}\frac{dk^+d^2k_\perp}{2k^+(2\pi)^3} \equiv dI^{vac} \Leftrightarrow k^+\frac{dI^{vac}}{dk^+d^2k_\perp} = \frac{\overline{|M_{vac}|^2}}{2(2\pi)^3} \tag{3.71}$$

The average square amplitude is just:

$$\overline{|M_{vac}|^2} = \frac{1}{d_A}\sum_{b,B}|M_{vac}|^2 = \frac{1}{N_c}\frac{4g^2}{k_\perp^2}\mathrm{Tr}\left[T^bT^b\right] = \frac{C_F 4g^2}{k_\perp^2} \tag{3.72}$$

And so, we get:

$$k^+\frac{dI^{vac}}{dk^+d^2k_\perp} = C_F\frac{4g^2}{2(2\pi)^2}\frac{1}{k_\perp^2} = C_F\frac{\alpha_s}{\pi^2}\frac{1}{k_\perp^2} \tag{3.73}$$

This is the leading $1/z$ contribution to the vacuum radiation spectrum. We can see from here that we get the leading part of vacuum splitting function for $z = 0$:

$$P_{q\leftarrow g}(z \simeq 0) \rightarrow 2C_F \tag{3.74}$$

Now that the easiest processes are done, we can pass to diagram 3.11(a) but with one scattering (fig. 3.14), so we can generalize to n scatterings like we have done in previous sections.

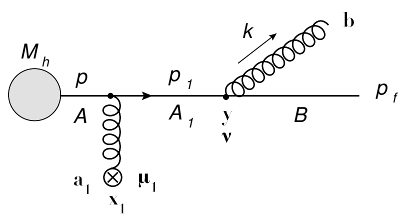


Figure 3.14: Diagram for quark propagation inside the medium with one scattering.

The S-matrix is given by:

$$\mathcal{S}_q^{(1)}(p_f,k) = \int d^4y \frac{d^4p_1}{(2\pi)^4} d^4x_1 \frac{d^4p}{(2\pi)^4} d^4x_0 \bar{u}(p_f) ig\gamma^\nu T^b_{BA_1}\epsilon^{*b}_\nu(k) \frac{i\not{p}_1}{p_1^2+\epsilon} ig\gamma^{\mu_1} T^{a_1}_{A_1A} A^{a_1}_{\mu_1}(x_1^+, \mathbf{x_1}_\perp) \frac{i\not{p}}{p^2+i\epsilon} \cdot$$
$$\cdot M_h(x_0) \mathrm{e}^{iy\cdot(p_f+k-p_1)} \mathrm{e}^{ix_1\cdot(p_1-p)} \mathrm{e}^{ix_0\cdot p} =$$
$$= \int dx_1^+ d^2\mathbf{x_1}_\perp \frac{dp^-}{2\pi} \frac{d^2\mathbf{p}_\perp}{(2\pi)^2} \bar{u}(p_f) ig\gamma^\nu T^b_{BA_1} \epsilon^{*b}_\nu(k) \frac{i(\not{p}_f+\not{k})}{2p_f\cdot k} ig\gamma^\mu T^{a_1}_{A_1A} A^{a_1}_{\mu_1}(x_1^+, \mathbf{x_1}_\perp) \cdot$$
$$\cdot \left.\frac{i\not{p}}{p^2+i\epsilon}\right|_{p^+=(p_f+k)^+} M_h(p_f+k) \mathrm{e}^{ix_1^+(p_f+k)^- - i\mathbf{x_1}_\perp\cdot(\mathbf{p_f}+\mathbf{k})_\perp} \mathrm{e}^{-ix_1^+p^- + i\mathbf{x_1}_\perp\cdot\mathbf{p}_\perp} \tag{3.75}$$

Applying the same approximations than before, we obtain:

$$\mathcal{S}_q^{(1)}(p_f,k) \simeq -2g \int dx_1^+ d^2\mathbf{x_1}_\perp T^b_{BA_1} \frac{\boldsymbol{\epsilon}^{*b}_\perp\cdot\mathbf{k}_\perp}{k_\perp^2} igA^-(x_1^+, \mathbf{x_1}_\perp)\theta(x_1^+)\delta^2(\mathbf{x_1}_\perp)\mathcal{S}_{el}(p_f+k, p_f) \cdot$$
$$\cdot \mathrm{e}^{ix_1^+(p_f+k)^- - i\mathbf{x_1}_\perp\cdot(\mathbf{p_f}+\mathbf{k})} =$$
$$= -2g \int dx_1^+ T^b_{BA_1} \frac{\boldsymbol{\epsilon}^{*b}_\perp\cdot\mathbf{k}_\perp}{k_\perp^2} igA^-(x_1^+, \mathbf{0})\theta(x_1^+)\mathcal{S}_{el}(p_f+k,p_f)\mathrm{e}^{ix_1^+(p_f+k)^-} \simeq$$
$$\simeq -2g \int dx_1^+ T^b_{BA_1} \frac{\boldsymbol{\epsilon}^{*b}_\perp\cdot\mathbf{k}_\perp}{k_\perp^2} igA^-(x_1^+, \mathbf{0})\theta(x_1^+)\mathcal{S}_{el}(p_f+k,p_f) \tag{3.76}$$

For two scatterings (fig. 3.15)

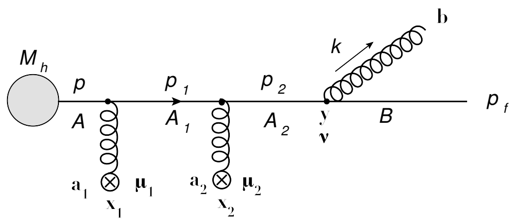


Figure 3.15: Diagram for quark propagation inside the medium with two scatterings.

$$\mathcal{S}_q^{(2)}(p_f,k) = \int d^4y \frac{d^4p_2}{(2\pi)^4} d^4x_2 \frac{d^4p_1}{(2\pi)^4} d^4x_1 \frac{d^4p}{(2\pi)^4} d^4x_0 \mathrm{e}^{iy\cdot(p_f+k-p_2)} \mathrm{e}^{ix_2\cdot(p_2-p_1)} \mathrm{e}^{ix_1\cdot(p_1-p)} \mathrm{e}^{ix_0\cdot p} \bar{u}(p_f) \cdot$$
$$\cdot ig\not{\epsilon}^{*b}(k) \frac{i\not{p}_2}{p_2^2+i\epsilon} ig\not{A}(x_2) \frac{i\not{p}_1}{p_1^2+i\epsilon} ig\not{A}(x_1) \frac{i\not{p}}{p^2+i\epsilon} M_h(x_0) \simeq$$
$$\simeq -2g \int dx_2^+ dx_1^+ \mathrm{e}^{ix_2^+(p_f+k)^-} \theta(x_2-x_1)^+ \theta(x_1)^+ T^b_{BA_2} igA^-(x_2^+, \mathbf{0}) igA^-(x_1^+, \mathbf{0}) \cdot$$
$$\cdot \mathcal{S}_{el}(pf+k, pf) \frac{\boldsymbol{\epsilon}^{*b}_\perp\cdot\mathbf{k}_\perp}{k_\perp^2} \tag{3.77}$$

Generalizing to n scatterings (diagram 3.16):

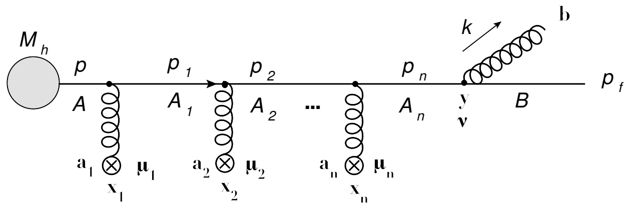


Figure 3.16: Diagram for quark propagation inside the medium with n scatterings.

$$
\begin{aligned}
\mathcal{S}_q^{(n)} = &-2g\int dx_n^+\dots dx_1^+ \mathrm{e}^{ix_n^+(p_f+k)^-}\theta(x_n-x_{n-1})^+\theta(x_{n-1}-x_{n-2})\dots\theta(x_2-x_1)\theta(x_1)\frac{\boldsymbol{\epsilon}_\perp^{*b}\cdot\mathbf{k}_\perp}{k_\perp^2}\cdot\\
&\cdot T^b_{BA_N} igA^-(x_n^+,\mathbf{0})\dots igA^-(x_1^+,\mathbf{0})\mathcal{S}_{el}(p_f+k,p_f)\simeq\\
&\simeq -2g\frac{\boldsymbol{\epsilon}_\perp^{*b}\cdot\mathbf{k}_\perp}{k_\perp^2}T^b_{BA_N}\frac{1}{n!}\mathcal{P}\left[\int dx^+A^-(x+,\mathbf{0})\right]^n\mathcal{S}_{el}(p_f+k,p_f)
\end{aligned}
\tag{3.78}
$$

and summing over scattering centres, we will find the expression for the process of medium-induced gluon radiation when the gluon is radiated outside the medium:

$$
\mathcal{S}_q^{rad} = \sum_{n=0}^{\infty}\mathcal{S}_q^{(n)} \simeq -2gT^b_{BA_N}\frac{\boldsymbol{\epsilon}_\perp^{*b}\cdot\mathbf{k}_\perp}{k_\perp^2}W_{A_nA_1}(\mathbf{0}_\perp;x_1^+,x_n^+)\mathcal{S}_{el}(p_f+k,p_f) \tag{3.79}
$$

Now we can pass to the next diagram where the gluon is radiated inside the medium (fig. 3.11(b)). For one scattering, we have the figure 3.17 and the $\mathcal{S}$-matrix (3.80):

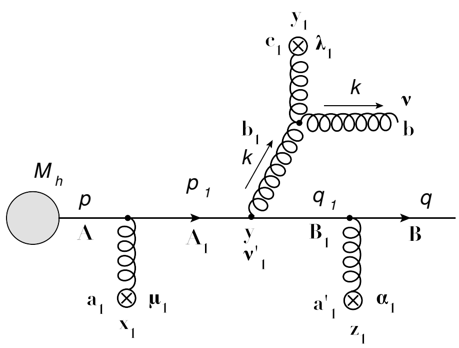


Figure 3.17: Diagram for gluon propagation inside the medium with one scatterings.

$$\begin{aligned}
\mathcal{S}_g^{(1)}(q,k) = &\int d^4z_1 \frac{d^4q_1}{(2\pi)^4} d^4y \frac{d^4p_1}{(2\pi)^4} d^4x_0 d^4y_1 \frac{d^4k_1}{(2\pi)^4} \mathrm{e}^{iz_1\cdot(q-q_1)} \cdot \mathrm{e}^{iy\cdot(q_1+k_1-p_1)} \mathrm{e}^{ix_1\cdot(p_q-p)} \mathrm{e}^{ix_0\cdot p} \mathrm{e}^{iy_1\cdot(k-k_1)} \cdot \\
&\cdot \bar{u}(q) T^{a_1'}_{BB_1} ig \not{A}^{a'}(z_1^+, \mathbf{x_1}_\perp) \frac{i\not{q}_1}{q_1^2+i\epsilon} ig T^{b_1}_{B_1A_1} \gamma^{\nu_1'} \frac{i\not{p}_1}{p_1^2+i\epsilon} ig \not{A}^{a_1}(x_1^+, \mathbf{x_1}_\perp) \frac{i\not{p}}{p^2+i\epsilon} M_h(x_0) \cdot \\
&\cdot \epsilon_\nu^{*b}(k) V^{bb_1c_1}_{\nu\nu_1\lambda_1}(-k,k,0) A^{c_1}_{\lambda_1}(y_1^+, \mathbf{y_1}_\perp) \frac{-ig_{\nu_1\nu_1'}}{k_1^2+i\epsilon} \simeq \\
\simeq &\int d^4z_1 \frac{d^4q_1}{(2\pi)^4} d^4x_1 \frac{d^4p}{(2\pi)^4} d^4y_1 \frac{d^4k_1}{(2\pi)^4} \mathrm{e}^{iz_1\cdot(q-q_1)} \mathrm{e}^{ix_1\cdot(q_1+k_1-p)} \mathrm{e}^{iy_1\cdot(k-k_1)} \bar{u}(q) ig \not{A}_{BB_1}(z_1^+, \mathbf{z_1}_\perp) \cdot \\
&\cdot \frac{i\not{q}_1}{q_1^2+i\epsilon} ig T^{b_1}_{B_1A_1} \not{\epsilon}^*(k) \frac{i(\not{q}_1+\not{k}_1)}{(q_1+k_1)^2} ig \not{A}_{A_1A}(x_1^+, \mathbf{x_1}_\perp) \frac{i\not{p}}{p^2+i\epsilon} g f^{bb_ac_a} A^{c_1-}(y_1^+, \mathbf{y_1}_\perp) \cdot \\
&\cdot \frac{i}{k_1^- - (k_{1_\perp}^2/2k_1^+ - i\epsilon)} \mathcal{S}_{el}(p)
\end{aligned} \tag{3.80}$$

where we have used the relation between the metric and polarization (A.24).

As for the colour matrices, the adjoint colour matrix is defined as:

$$(T_G^a)^{bc} = -if^{abc} \tag{3.81}$$

which means that we can write the fields in the adjoint representation:

$$\begin{gathered}
-if^{abc}A^{c-}(y) = (T_G^a)^{bc} Ac-(y) = (T_G^c)^{ab} A^{c-}(y) \Leftrightarrow \\
\Leftrightarrow f^{abc}A^{c-}(y) = i\,(T_G^a)^{bc} A^{c-}(y) \equiv iA^-_{bc}(y)
\end{gathered} \tag{3.82}$$

To know to what representation we refer to, one just has to pay attention to the indices: if they are capital letters, then the field is in the fundamental or anti-fundamental representation; if they are small letters, then the field is in the adjoint representation.

With this, and after some algebraic manipulations, we get the following result for the $\mathcal{S}$-matrix:

$$\begin{aligned}
\mathcal{S}_g^{(1)}(q,k) \simeq 2gi &\int dx_1^+ dy^+ dz_1^+ dy_1^+ d\mathbf{y_1}_\perp \mathcal{S}_{el}(q+k,q) \frac{\boldsymbol{\epsilon}^*_\perp \cdot \mathbf{k}_\perp}{2k^+} \theta(z_1-y)^+ \theta(y-x_1)^+ \theta(x_1)^+ \theta(y_1-y)^+ \cdot \\
&\cdot igA^-_{BB_1}(x_1^+, \mathbf{0}_\perp) T^{b_1}_{B_1A_1} igA^-_{bb_1}(y_1^+, \mathbf{y_1}_\perp) G_0(y^+, \mathbf{0}_\perp; y_1^+, \mathbf{y_1}_\perp | k^+) \mathrm{e}^{iz_1^+q^- + iy_1^+k^- - i\mathbf{y_1}_\perp\cdot\mathbf{k}_\perp}
\end{aligned} \tag{3.83}$$

From quantum mechanics we know that the momentum operator can be written in the coordinate basis as:

$$\mathbf{p} = -ih\boldsymbol{\nabla} \Rightarrow \mathbf{k}_\perp \equiv -i\frac{\partial}{\partial \mathbf{y}_\perp} \tag{3.84}$$

Using this relation, the final expression for one scattering should be:

$$M_g^{(1)}(q,k) \simeq \frac{g}{k^+}\int dx_1^+ dy_1^+ dz_1^+ dy^+ d\mathbf{y_1}_\perp \theta(z_1-y)^+\theta(y-x_1)^+\theta(x_1)^+\theta(y_1-y)^+ \mathrm{e}^{iz_1^+q^- + iy_1^+k^- - i\mathbf{y_1}_\perp\cdot\mathbf{k}_\perp}\cdot$$
$$\cdot\, igA^-_{BB_1}(z_1^+,\mathbf{0}_\perp)T^{b_1}_{B_1A_1} igA^-_{A_1A}(x_1^+,\mathbf{0}_\perp)\boldsymbol{\epsilon}_\perp\cdot\frac{\partial}{\partial\mathbf{y}_\perp}G_0(y^+,\mathbf{0}_\perp;y_1^+,\mathbf{y_1}_\perp|k^+) \tag{3.85}$$

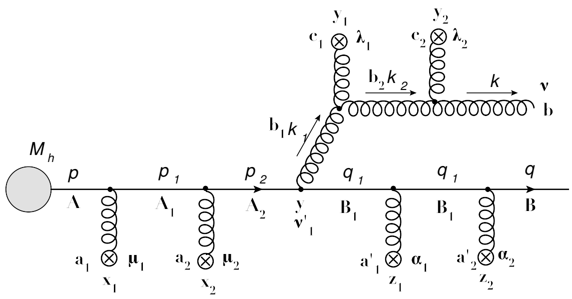


Figure 3.18: Diagram for gluon propagation inside the medium with two scatterings.

For two scatterings, like shown in figure 3.18 we have:

$$\mathcal{S}_g^{(2)}(q,k) = \int d^4z_1\, d^4z_2\, d^4y\, d^4x_2\, d^4x_1\, d^4y_2\, d^4y_1 \frac{d^4q_2}{(2\pi)^4}\frac{d^4q_1}{(2\pi)^4}\frac{d^4p_2}{(2\pi)^4}\frac{d^4p_1}{(2\pi)^4}\frac{d^4p}{(2\pi)^4}\frac{d^4k_2}{(2\pi)^4}\frac{d^4k_1}{(2\pi)^4}d^4x_0\cdot$$
$$\cdot\, \mathrm{e}^{i\cdot(q-q_2)+iz_1\cdot(q_2-q_1)+iy\cdot(q_1+k_1-p_2)+ix_2(p_2-p_1)}\mathrm{e}^{ix_0\cdot p}\mathrm{e}^{iy_2\cdot(k-k_2)+iy_1\cdot(k_2-k_1)}\bar{u}(q)\cdot$$
$$\cdot\, ig\not{A}_{BB_2}(z_2)\frac{i\not{q}_2}{q_2^2+i\epsilon}ig\not{A}_{B_2B_1}(z_1)\frac{i\not{q}_1}{q_1^+ + i\epsilon}ig\gamma^{\nu_1'}T^{b_1}_{B_1A_2}\frac{i\not{p}_2}{p_2^2+i\epsilon}ig\not{A}_{A_2A_1}(x_2)\frac{i\not{p}_1}{p_1^2+i\epsilon}\cdot$$
$$\cdot\, ig\not{A}_{A_1A}(x_1)\frac{i\not{p}}{p^2+i\epsilon}\epsilon^*_\nu(k)gf^{bb_2c_2}A^{c_2}_{\lambda_2}(y_2)g^{\nu\nu_2}(-2k)^{\lambda_2}\frac{-ig_{\nu_2\nu_2'}}{k_2^2+i\epsilon}gf^{b_2b_1c_1}A^{c_1}_{\lambda_1}(y_1)\cdot$$
$$\cdot\, g^{\nu_2'\nu_1}(-2k)^{\lambda_1}\frac{-ig_{\nu_1\nu_1'}}{k_1^2+i\epsilon}M_h(x_0) \tag{3.86}$$

After all possible integrations and simplifications, we will have as result the following:

$$\mathcal{S}_g^{(2)}(q,k) = \frac{g}{k^+}\boldsymbol{\epsilon}_\perp\frac{\partial}{\partial\mathbf{y}_\perp}\int dz_2^+ dz_1^+ dy^+ dx_2^+ dx_1^+ dy_2^+ dy_1^+ d\mathbf{y_2}_\perp d\mathbf{y_1}_\perp \mathrm{e}^{iz_2^+q^- + iy_2^+k^- - i\mathbf{y_2}_\perp\cdot\mathbf{k}_\perp}\mathcal{S}_{el}(q+k,q)\cdot$$
$$\cdot\, igA^-_{BB_2}(z_2^+,\mathbf{0}_\perp)igA^-_{B_2B_1}(z_1^+,\mathbf{0}_\perp)T^{b_1}_{B_1A_2}igA^-_{A_2A_1}(x_2^+,\mathbf{0}_\perp)igA^-_{A_1A}(x_1^+,\mathbf{x_1}_\perp)\cdot$$
$$\cdot\, igA^-_{bb_2}(y_2^+,\mathbf{y_2}_\perp)ifA^-_{b_2b_1}(y_1^+,\mathbf{y_1}_\perp)\theta(z_2-z_1)^+\theta(z_1-y)^+\theta(y-x_2)^+\theta(x_2-x_1)^+\theta(x_1)^+\cdot$$
$$\cdot\, \theta(y_2-y_1)^+\theta(y_1-y)^+G_0(y_1^+,\mathbf{y_1}_\perp;y_2^+,\mathbf{y_2}_\perp)G_0(y^+,\mathbf{0}_\perp;y_1^+,\mathbf{y_1}_\perp) \tag{3.87}$$

It is easy to see that from here, the radiation amplitude for the whole process 3.19(a) should be:

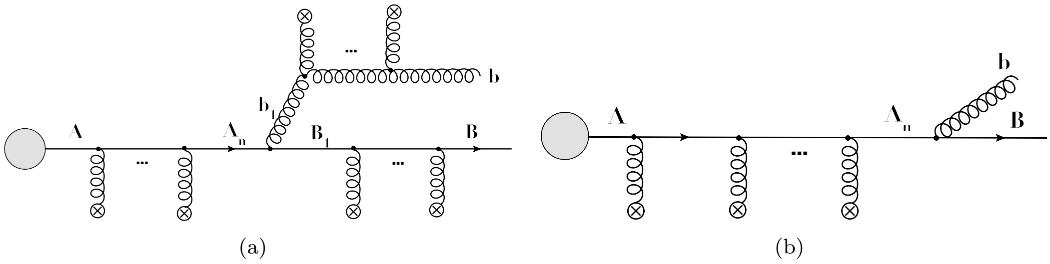


Figure 3.19: Diagrams for gluon (a) and quark (b) propagation inside the medium with n scatterings

$$M_g^{rad} \simeq \frac{g}{k^+}\int_{x_0^+}^{L^+} dy^+ d^2\mathbf{y}_\perp \mathrm{e}^{-i\mathbf{y}_\perp\cdot\mathbf{k}_\perp} W_{BB_1}(\mathbf{0}_\perp; y^+, L^+) T^{b_1}_{B_1 A_n} W_{A_n A}(\mathbf{0}_\perp; x_0^+, y^+) \boldsymbol{\epsilon}_\perp^{*\boldsymbol{b_1}} \cdot \\ \cdot \frac{\partial}{\partial \mathbf{y}_\perp} G_{bb_1}(y^+, \mathbf{0}_\perp; L^+, \mathbf{y}_\perp) \tag{3.88}$$

where we made explicit the limits of the medium: start at x_0^+ and end at L^+. Doing the same for the quark radiation (fig. 3.19(b)):

$$M_q^{(rad)} \simeq -2gT^b_{BA_n}\frac{\boldsymbol{\epsilon}_\perp^{*\boldsymbol{b}}\cdot\mathbf{k}_\perp}{k_\perp^2} W_{A_n A_1}(\mathbf{0}_\perp; x_0^+, L^+) \tag{3.89}$$

The total amplitude for the medium-induced gluon radiation will be the sum of these two components:

$$M^{rad} = M_q^{rad} + M_g^{rad} \tag{3.90}$$

which means:

$$\overline{|M^{rad}|^2} = \overline{\left|M_q^{rad}\right|^2} + \overline{\left|M_g^{rad}\right|^2} + \overline{\left(M_q^{rad}\right)\left(M_g^{rad}\right)^\dagger} + \overline{\left(M_q^{rad}\right)^\dagger\left(M_g^{rad}\right)} \tag{3.91}$$

We will compute all terms separately:

- $\overline{\left|M_q^{rad}\right|^2}$

$$\overline{\left|M_q^{rad}\right|^2} = \frac{1}{2d_A}\frac{4g^2}{k_\perp^2} T^b_{BA_n} T^b_{\bar{A}_n B}\left\langle W_{A_n A}(\mathbf{0}_\perp; x_0^+, L^+) W^\dagger_{A\bar{A}_n}(\mathbf{0}_\perp; x_0^+, L^+)\right\rangle = \\ = \frac{2g^2}{N_c}\frac{C_F}{k_\perp^2}\delta_{\bar{A}_n A_n}\left\langle W_{A_n A}(\mathbf{0}_\perp; x_0^+, L^+) W^\dagger_{A\bar{A}_n}(\mathbf{0}_\perp; x_0^+, L^+)\right\rangle = \\ = \frac{g^2}{k_\perp^2}\frac{2C_F}{N_c}\mathrm{Tr}\left\langle \mathbf{1}\right\rangle = \frac{g^2}{k_\perp^2}2C_F \tag{3.92}$$

- $\overline{\left|M_g^{rad}\right|^2}$

There exist two possible cases: $y^+ < y_c^+$ or $y^+ > y_c^+$, where $(y^+, \mathbf{y}_\perp)$ corresponds to coordinates in the amplitude and $(y_c^+, \mathbf{y}_\perp{}')$ corresponds to coordinates in the complex conjugate amplitude.

Analyzing the first case, $y^+ < y_c^+$, we have:

$$\begin{aligned}\overline{|M_g^{rad}|^2}^{(1)} &= \frac{g^2}{2(k^+)^2 d_A} \int_{x_0^+}^{L^+} dy_c^+ \int_{y_c^+}^{L^+} dy^+ \int d\mathbf{y}'_\perp d\mathbf{y}_\perp \mathrm{e}^{-i\mathbf{k}_\perp\cdot(\mathbf{y}_\perp - \mathbf{y}'_\perp)} \Big\langle W_{BB_1}(\mathbf{0}_\perp; y^+, L^+) T^{b_1}_{B_1 A_n} \cdot \\ &\cdot W_{A_n A}(\mathbf{0}_\perp; x_0^+, y^+) \boldsymbol{\epsilon}^*_\perp \cdot \frac{\partial}{\partial \mathbf{y}_\perp} G_{bb_1}(y^+, \mathbf{0}_\perp; L^+, \mathbf{y}_\perp) \boldsymbol{\epsilon}_\perp \cdot \frac{\partial}{\partial \mathbf{y}'_\perp} G^\dagger_{\bar{b}_1 b}(y_c^+, \mathbf{0}_\perp; L^+, \mathbf{y}_\perp{}') \cdot \\ &\cdot W^\dagger_{A\bar{A}_n}(\mathbf{0}_\perp; x_0^+, y_c^+) T^{\bar{b}_1}_{\bar{A}_n \bar{B}_1} W^\dagger_{\bar{B}_1 B}(\mathbf{0}_\perp; y_c^+, L^+) \Big\rangle\end{aligned} \tag{3.93}$$

Drawing the path of all Wilson lines like in fig. 3.20 we can simplify the above structure:

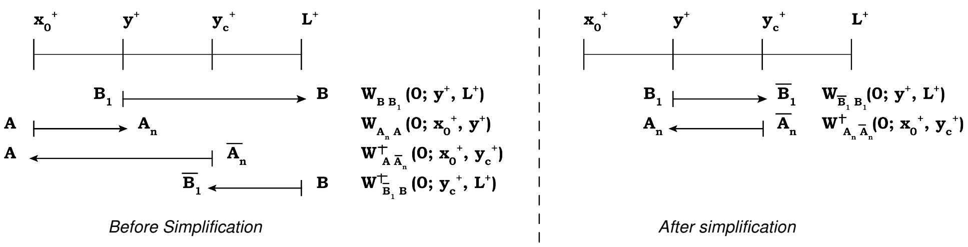


Figure 3.20: Schematic path of the Wilson Lines

Furthermore, we know that the adjoint Wilson line is defined as:

$$W_A^{ab}(\mathbf{x}_\perp; x^+, y^+) = 2\mathrm{Tr}\left[W_F^\dagger(\mathbf{x}_\perp; x^+, y^+) T^a W_F(\mathbf{x}_\perp; x^+, y^+) T^b\right] \tag{3.94}$$

Due the properties of the colour matrices:

$$(T^a)^\dagger = \left[(T^a)^*\right]^T = \left[(T^a)^T\right]^* = T^a \Rightarrow (T^a)^T = (T^a)^* \tag{3.95}$$

the adjoint Wilson line, have the following useful properties that can be used in our calculations:

$$\left(W_A^{ab}\right)^\dagger = 2\mathrm{Tr}\left[T^b W_F^\dagger T^a W_F\right] = \left(W_A^{ab}\right) \tag{3.96}$$

$$\begin{aligned}\left(W_A^{ab}\right)^* &= 2\mathrm{Tr}\left[\left(W_F^\dagger\right)^* (T^a)^* (W_F)^* \left(T^b\right)^*\right] = 2\mathrm{Tr}\left[(W_F)^T (T^a)^T \left(W_F^\dagger\right)^T \left(T^b\right)^T\right] = \\ &= 2\mathrm{Tr}\left[T^b W_F^\dagger T^a W_F\right] = W_A^{ab}\end{aligned} \tag{3.97}$$

$$\Rightarrow W_A^{ba} \equiv \left(W_A^{ab}\right)^T = \left(W_A^{ab}\right)^* = \left(W_A^{ab}\right)^\dagger = W_A^{ab} \tag{3.98}$$

Returning to our amplitude, after the algebraic manipulations mentioned above and using the properties of the adjoint Wilson line, we arrive at:

$$
\begin{aligned}
\overline{\left|M_g^{rad}\right|^2}^{(1)} &= \frac{g^2}{2N_c(k^+)^2}\int_{x_0^+}^{L^+} dy_c^+ \int_{x_0^+}^{y_c^+} dy^+ \int d\mathbf{y}_\perp \mathbf{y}'_\perp \mathrm{e}^{-i\mathbf{k}_\perp\cdot(\mathbf{y}_\perp-\mathbf{y}'_\perp)} \Big\langle W^A_{b_1\bar{b}_1}\cdot \\
&\cdot(\mathbf{0}_\perp; y^+, y_c^+)\frac{\partial}{\partial \mathbf{y}_\perp} G_{bb_1}(y^+,\mathbf{0}_\perp; L^+, \mathbf{y}_\perp)\frac{\partial}{\partial \mathbf{y}'_\perp} G^\dagger_{\bar{b}_1 b}(y_c^+, \mathbf{o}_\perp; l^+, \mathbf{y}'_\perp)\Big\rangle
\end{aligned} \tag{3.99}
$$

Drawing the path of these propagators, like shown in fig. 3.21, in order to close the trace, the adjoint Wilson line must be conjugated. Also, the G_{bb_1} can be cut in two pieces, and we may write the following composition of propagators with $x^+ < z^+$:

$$
G_{bb_1}(y^+,\mathbf{0}_\perp; L^+,\mathbf{y}_\perp) = \int d\mathbf{z}_\perp G_{bd}(y_c^+,\mathbf{z}_\perp; L^+,\mathbf{y}_\perp) G_{db_1}(y^+,\mathbf{0}_\perp; y_x^+,\mathbf{z}_\perp) \tag{3.100}
$$

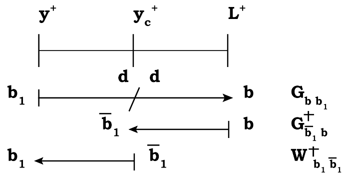


Figure 3.21: Schematic path of the adjoint Wilson line and Green's functions

This allows to write the amplitude as:

$$
\begin{aligned}
\overline{\left|M_g^{rad}\right|^2}^{(1)} &= \frac{g^2}{2N_c(k^+)^2}\int_{x_0^+}^{L^+} dy_c^+ \int_{x_0^+}^{y_c^+} dy^+ \int d\mathbf{y}_\perp d\mathbf{y}'_\perp \mathrm{e}^{-i\mathbf{k}_\perp\cdot(\mathbf{y}_\perp-\mathbf{y}'_\perp)}\cdot \\
&\cdot\left\langle \left[G(y^+,\mathbf{0}_\perp; y_c^+,\mathbf{z}_\perp) W_A^\dagger(\mathbf{0}_\perp; y^+, y_c^+)\right]_{d\bar{b}_1}\right\rangle \frac{\partial}{\partial \mathbf{y}_\perp}\frac{\partial}{\partial \mathbf{y}'_\perp}\cdot \\
&\cdot\left\langle \left[G^\dagger(y_c^+, \mathbf{y}_\perp^+ = \mathbf{0}; L^+, \mathbf{y}_\perp) G(y_c^+,\mathbf{z}_\perp; L^+,\mathbf{y}_\perp)\right]_{\bar{b}_1 d}\right\rangle
\end{aligned} \tag{3.101}
$$

Using equation (3.52), we will end at:

$$
\begin{aligned}
\overline{\left|M_g^{rad}\right|^2}^{(1)} &= \frac{g^2}{2(k^+)^2}\frac{1}{N_c(N_c^2-1)}\int_{x_0^+}^{L^+} dy_c^+ \int_{x_0^+}^{y_c^+} dy^+ \int d\mathbf{y}_\perp d\mathbf{y}'_\perp d\mathbf{z}_\perp \mathrm{e}^{-i\mathbf{k}_\perp\cdot(\mathbf{y}_\perp-\mathbf{y}'_\perp)}\cdot \\
&\cdot \mathrm{Tr}\left\langle G(y^+,\mathbf{0}_\perp; y_c^+,\mathbf{z}_\perp) W_A^\dagger(\mathbf{0}_\perp; y^+, y_c^+)\right\rangle \frac{\partial}{\partial \mathbf{y}_\perp}\frac{\partial}{\partial \mathbf{y}'_\perp}\cdot \\
&\cdot \mathrm{Tr}\left\langle G^\dagger(y_c^+, \mathbf{y}'_\perp = \mathbf{0}; L^+, \mathbf{y}_\perp) G(y_c^+,\mathbf{z}_\perp; L^+,\mathbf{y}_\perp)\right\rangle = \\
&= \frac{g^2}{(k^+)^2} C_F \int_{x_0^+}^{L^+} dy_c^+ \int_{x_0^+}^{y_c^+} dy^+ \int d\mathbf{z}_\perp \mathrm{e}^{-i\mathbf{k}_\perp\cdot\mathbf{z}_\perp} \mathrm{e}^{-\frac{1}{2}\int_{y_c^+}^{L^+} d\xi n(\xi)\sigma(\mathbf{z}_\perp)}\cdot \\
&\cdot \mathcal{K}(y^+,\mathbf{0}_\perp; y_c^+,\mathbf{z}_\perp)
\end{aligned} \tag{3.102}
$$

Now, for the second case ($y^+ > y_c^+$), if we apply the same procedure as above, the result will be:

$$
\begin{aligned}
\overline{\left|M_g^{rad}\right|^2}^{(2)} &= \frac{g^2}{2(k^+)^2}\frac{1}{N_c(N_c^2-1)}\int_{x_0^+}^{L^+} dy^+ \int_{x_0^+}^{y^+} dy_c^+ \int d\mathbf{y}_\perp d\mathbf{y}'_\perp d\mathbf{z}_\perp \mathrm{e}^{-i\mathbf{k}_\perp\cdot(\mathbf{y}_\perp-\mathbf{y}'_\perp)}\cdot \\
&\cdot \mathrm{Tr}\left\langle W^A(\mathbf{0}_\perp; y_c^+, y^+)G^\dagger(y_c^+.\mathbf{0}_\perp; y^+, \mathbf{z}_\perp)\right\rangle \frac{\partial}{\partial\mathbf{y}_\perp}\frac{\partial}{\partial\mathbf{y}'_\perp}\cdot \\
&\cdot \mathrm{Tr}\left\langle G^\dagger(y^+, \mathbf{z}_\perp; L^+, \mathbf{y}'_\perp)G(y^+, \mathbf{y}'_\perp=\mathbf{0}; L^+, \mathbf{y}_\perp)\right\rangle
\end{aligned}
\tag{3.103}
$$

When comparing with $\overline{\left|M_g^{rad}\right|^2}^{(1)}$, equation (3.102), we must exchange $y^+ \leftrightarrow y_c^+$ and $\mathbf{y}_\perp \leftrightarrow \mathbf{y}'_\perp$:

$$
\begin{aligned}
\overline{\left|M_g^{rad}\right|^2}^{(2)} &= \frac{g^2}{2(k^+)^2}\frac{1}{N_c(N_c^2-1)}\int_{x_0^+}^{L^+} dy_c^+ \int_{x_0^+}^{L^+} dy^+ \int d\mathbf{y}_\perp d\mathbf{y}'_\perp d\mathbf{z}_\perp \mathrm{e}^{i\mathbf{k}_\perp\cdot(\mathbf{y}_\perp-\mathbf{y}'_\perp)}\cdot \\
&\cdot \mathrm{Tr}\left\langle W^A(\mathbf{0}_\perp; y^+, y_c^+)G^\dagger(y^+, \mathbf{0}_\perp; y_c^+, \mathbf{z}_\perp)\right\rangle \frac{\partial}{\partial\mathbf{y}_\perp}\frac{\partial}{\partial\mathbf{y}'_\perp}\cdot \\
&\cdot \mathrm{Tr}\left\langle G^\dagger(y_c^+, \mathbf{z}_\perp; L^+, \mathbf{y}_\perp)G(y_c^+, \mathbf{0}_\perp; L^+, \mathbf{y}'_\perp)\right\rangle
\end{aligned}
\tag{3.104}
$$

It is clear from here that:

$$
\overline{\left|M_g^{rad}\right|^2}^{(2)} = \left(\overline{\left|M_g^{rad}\right|^2}^{(1)}\right)^\dagger \tag{3.105}
$$

But $z + z^* = 2\mathrm{Re}(z)$ and so, our final expression for $\overline{\left|M_g^{rad}\right|^2}$ will be:

$$
\begin{aligned}
\overline{\left|M_g^{rad}\right|^2} = \frac{g^2}{(k^+)^2}C_F 2\mathrm{Re}\Bigg\{&\int_{x_0^+}^{L^+} d_c^+ \int_{x_0^+}^{y_c^+} dy^+ \int d\mathbf{y}'_\perp \mathrm{e}^{-i\mathbf{k}_\perp\cdot\mathbf{y}'_\perp}\mathrm{e}^{-\frac{1}{2}\int_{y_c^+}^{L^+} d\xi n(\xi)\sigma(\mathbf{z}_\perp)}\cdot \\
&\cdot\frac{\partial}{\partial\mathbf{y}_\perp}\frac{\partial}{\partial\mathbf{y}'_\perp}\mathcal{K}(y^+, \mathbf{y}_\perp=\mathbf{0}; y_c^+, \mathbf{y}_\perp)\Bigg\}
\end{aligned}
\tag{3.106}
$$

- $\overline{\left(M_q^{rad}\right)\left(M_g^{rad}\right)^\dagger} + \overline{\left(M_q^{rad}\right)^\dagger\left(M_g^{rad}\right)}$

For these last two terms, we can also write them as:

$$
\begin{aligned}
2\mathrm{Re}\left[\left(M_Q^{rad}\right)^\dagger\ \left(M_g^{rad}\right)\right] &= 2\mathrm{Re}\Bigg\{\frac{-1}{2d_A}\frac{2g^2}{k^+}\frac{1}{\mathbf{k}_\perp}\int_{x_0^+}^{L^+} dy^+ d\mathbf{y}_\perp \mathrm{e}^{-i\mathbf{y}_\perp\cdot\mathbf{k}_\perp}\Big\langle W^\dagger_{A\bar{A_n}}(\mathbf{0}_\perp; x_0^+, L^+)T^b_{\bar{A_n}B}\cdot \\
&\cdot\ W_{BB_1}(\mathbf{0}_\perp; y^+, L^+)T^{b_1}_{B_1A_n}W_{A_nA}(\mathbf{0}_\perp; x_0^+, y^+)\frac{\partial}{\partial\mathbf{y}_\perp}G_{bb_1}(y^+, \mathbf{0}_\perp; L^+, \mathbf{y}_\perp)\Big\rangle\Bigg\} = \\
&= \frac{-2g^2}{k^+\mathbf{k}_\perp}C_R\ 2\mathrm{Re}\left\{\int_{x_0^+}^{L^+} dy^+ \int d\mathbf{y}_\perp \mathrm{e}^{-i\mathbf{k}_\perp\cdot\mathbf{y}_\perp}\mathcal{K}(y^+, \mathbf{0}_\perp; L^+, \mathbf{y}_\perp)\right\}
\end{aligned}
\tag{3.107}
$$

Joining all pieces, we will finally get the single inclusive distribution of gluons, emitted with energy $k^+ \equiv \omega$ and transverse momentum $\mathbf{k}_\perp$, from a parent parton with energy $E = \omega/x$:

$$
\begin{aligned}
\omega \frac{dI}{d\omega d^2k_\perp} &= \frac{\alpha_s}{(2\pi)^2}\frac{2C_F}{k^+}\mathrm{Re}\left\{\int_{x_0^+}^{L^+} dy^+ \int d\mathbf{y}_\perp \mathrm{e}^{-i\mathbf{k}_\perp\cdot\mathbf{y}_\perp}\left[\frac{1}{k^+}\int_{y^+}^{L^+}\int d\mathbf{y}'_\perp \mathrm{e}^{-\frac{1}{2}\int_{y^+}^{L^+} d\xi n(\xi)\sigma(\mathbf{y}_\perp)}\cdot\right.\right.\\
&\left.\left.\cdot \frac{\partial}{\partial \mathbf{y}_\perp}\frac{\partial}{\partial \mathbf{y}'_\perp}\mathcal{K}_\omega(y^+,\mathbf{y}'_\perp=\mathbf{0};y_c^+,\mathbf{y}_\perp) - 2\frac{\mathbf{k}_\perp}{k_\perp^2}\frac{\partial}{\partial \mathbf{y}'_\perp}\mathcal{K}_\omega(y^+,\mathbf{y}'_\perp=\mathbf{0}_\perp;L^+,\mathbf{y}_\perp)\right]\right\}+\\
&+\frac{\alpha_s}{(2\pi)^2}\frac{2C_F}{k_\perp^2}\equiv\\
&\equiv \omega\frac{dI^{med}}{d\omega d^2k_\perp}+\omega\frac{dI^{vac}}{d\omega d^2k_\perp}
\end{aligned}
\tag{3.108}
$$

The variables $\mathbf{y}_\perp$ and $\mathbf{y}'_\perp$ are distances between the positions of projectile components in the amplitude and complex amplitude. y^+ and y_c^+ are longitudinal coordinates integrated over the ordered longitudinal gluon emission points in the amplitude and complex amplitude, and C_R determines the coupling strength of the emitted gluon to the parent quark ($C_R = C_F$) or gluon ($C_R = C_A$).

It should be noted that the spectrum (3.108) is obtained by assuming small angle multiple scattering for both the projectile and the radiated gluon. This means, that its region of validity is limited to $x \ll 1$ and $|\mathbf{k}_\perp| \ll \omega$.

Also, all the medium dependence stems from the factor $n(\xi)\sigma(\mathbf{y}_\perp)$: $n(\xi)$ determines the density of scattering centres in the medium and the dipole cross section $\sigma(\mathbf{y}_\perp)$ contains configuration space information on the strength of a single elastic differential scattering cross section.

3.2.5 Medium Modified Splitting Functions

The cross section for producing a hadron h at a high transverse momentum $\mathbf{p}_\perp$ and rapidity y can be written as we already saw in chapter 1, in the factorized form:

$$
\frac{d\sigma^h}{d^2p_\perp dy} = \int_0^1 dx_1 \int_0^1 dx_2 \sum_{i,j} f_i(x_1,\mu^2) f_j(x_2,\mu^2)\frac{d\hat{\sigma}^{ij\to k}}{dt}\left(p_1,p_2,\alpha(\mu^2),Q^2/\mu^2\right) D_{k\to h}^{vac/med}(z,\mu^2)
\tag{3.109}
$$

In pp collisions, PDFs $f_p^i(x,Q^2)$ and vacuum FFs $D_{k\to h}^{vac}(z,\mu^2)$ are known from global fits to experimental data using the DGLAP evolution equations. In the nuclear case, PDFs are known from similar fits, but the FFs $D_{k\to h}^{med}(z,\mu^2)$ contain the information about the medium we want to study. So, we can apply this formalism studied so far, of medium-induced gluon radiation, to compute the medium modification of the FF and to characterized the medium properties, like was done in the BDMPS-Z/ASW model.

Because the theoretical description of the fragmentation of a high-$\mathbf{p}_\perp$ particle in the presence of a medium is not completely known from first principle calculations, some degree of modeling is needed. One possibility is to include all modifications of the FF into a modified splitting function in the DGLAP evolution equations like referred before. If we assume that the multiple gluon emission when the projectile rescatters with the nuclei are independent, then all medium effects can be included in a redefinition of the splitting function:

$$P^{tot}(z) = P^{vac}(z) + \Delta P(z,t) \tag{3.110}$$

where $\Delta P(z,t)$ is just taken from the medium-induced gluon radiation by comparing the leading contribution in the vacuum case:

$$\Delta P(z,t) \simeq \frac{(2\pi)^2}{\alpha_s}\omega\frac{dI^{med}}{d\omega d^2k_\perp} \tag{3.111}$$

Of course that this independence could be broken if non-trivial colour reconnections are present. The size of these corrections for the DGLAP kinematics has not been computed yet and so, it is of common use to assume (3.110).

In order to find a numerical expression for the medium-modified splitting function, we have to use some scheme of approximation to solve the medium part of the spectrum (3.108), since the exact solution of the path integral is unknown. The two approximations usually employed are:

- Opacity expansion: corresponds to expanding the spectrum in powers of opacity $n(\xi)\sigma(\mathbf{r})$. In this way, we have access to a small number of scattering centres.
- Multiple soft scattering or dipole approximation: corresponds to approximate the dipole cross section by its quadratic term:

$$n(\xi)\sigma(\mathbf{r}) \simeq \frac{1}{2}\hat{q}(\xi)\mathbf{r}^2 \tag{3.112}$$

  This allows to explore the multiple scattering regime in which the Brownian motion dominates.

We will use the second.

As for the study of the medium, there exist two classes of models:

- Static medium, in which $\hat{q}(\xi) = \hat{q}$ is a constant:

$$\hat{q} = \frac{\langle q_\perp^2 \rangle_{med}}{\lambda} \tag{3.113}$$

- Expanding medium, in which the density of scattering centres is expected to produce a dilution as:

$$\hat{q}(\xi) = \hat{q}_0 \left(\frac{\xi_0}{\xi}\right)^\alpha \tag{3.114}$$

  where α characterizes the speed of dilution.

In this thesis, we will only see the case of a static medium. Results for an expanding medium can be seen at [47, 51].

With these approximations the path integral of (3.108) is equivalent to that of a harmonic oscillator with mass ω and imaginary frequency Ω:

$$\mathcal{K}_{osc}(\mathbf{r}(x), x; \mathbf{r}(y), y) = \int \mathcal{D}\mathbf{r} \exp\left\{\frac{i\omega}{2}\int_x^y d\xi \left[\dot{\mathbf{r}}^2 - \frac{\hat{q(\xi)}\mathbf{r}^2}{2i\omega}\right]\right\} \tag{3.115}$$

Its solution is known and is given by [52]:

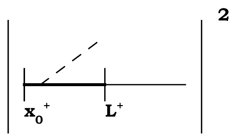
$$\mathcal{K}_{osc}(\mathbf{x}, x; \mathbf{y}, y) = \frac{A}{\pi i}\exp\left[iAB(\mathbf{y}^2 + \mathbf{x}^2) - 2iA\mathbf{y}\cdot\mathbf{x}\right] \tag{3.116}$$

with,

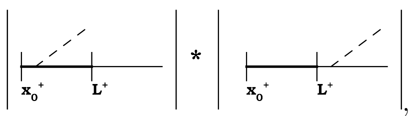
$$A = \frac{\mu\Omega}{2\sin\left[(y-x)\Omega\right]}, \qquad B = \cos\left[(y-x)\Omega\right], \qquad \Omega = \frac{(1-i)}{\sqrt{2}}\sqrt{\frac{q(\xi)}{\omega}} \tag{3.117}$$

It is usual to write the numerical expressions for the spectrum by splitting it according with the position of the radiation vertex being inside or outside the medium in the amplitude and/or complex conjugate amplitude:

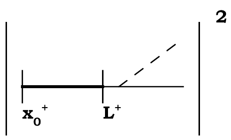
$$\omega\frac{dI}{d\omega d^2k_\perp} = \frac{2\alpha_s C_F}{(2\pi)^2}\left(I_4 + I_5 + I_6\right) \tag{3.118}$$

$I_4 =$ 

, the direct production term, (3.119a)

$I_5 = 2\mathrm{Re}$ 

, the destructive interference term, (3.119b)

$I_6 =$ 

, the hard vacuum radiation term, which is completely independent of any approximation scheme used. This is exactly the medium-independent contribution associated to the hard radiation off a nascent quark jet propagating in the vacuum.

(3.119c)

$$I_4 = \mathrm{Re}\left\{\frac{1}{\omega^2}\int_{x_0^+}^{L^+} dy^+ \int_{y^+}^{L^+} dy_c^+ \int d\mathbf{y}_\perp \mathrm{e}^{-i\mathbf{k}_\perp\cdot\mathbf{y}_\perp}\mathrm{e}^{-\frac{1}{4}(L^+-y^+)\hat{q}\mathbf{y}_\perp{}^2}\frac{\partial}{\partial\mathbf{y}_\perp}\frac{\partial}{\partial\mathbf{y}'_\perp}\cdot\right. \tag{3.120a}$$
$$\left.\cdot\,\mathcal{K}_{osc}(y^+,\mathbf{y}'_\perp=\mathbf{0};y_c^+,\mathbf{y}_\perp)\right\}$$
$$I_5 = \mathrm{Re}\left\{-2\frac{\mathbf{k}_\perp}{k_\perp^2}\frac{1}{\omega}\int_{x_0^+}^{L^+} dy^+ \int d\mathbf{y}_\perp\frac{\partial}{\partial\mathbf{y}_\perp}\mathcal{K}_{osc}(y^+,\mathbf{0}_\perp;L^+,\mathbf{y}_\perp)\right\} \tag{3.120b}$$
$$I_6 = \frac{1}{k_\perp^2} \tag{3.120c}$$

Integrating over $\mathbf{y}_\perp$ we get:

$$I_4 = \frac{1}{\omega^2}\mathrm{Re}\left\{\int_{x_0^+}^{L^+} dy^+ \int_{y^+}^{L^+} dy_c^+ \left[\frac{-4A_4^2D}{(D-iA_4B_4)^2}+\frac{iA_4^3B_4k_\perp^2}{(D-iA_4B_4)^3}\right]\exp\left[-\frac{k_\perp^2}{4(D-iA_4B_4)}\right]\right\} \tag{3.121a}$$
$$I_5 = \frac{2}{\omega}\mathrm{Re}\left\{\int_{x_0^+}^{L^+} dy^+ \frac{-i}{B_5^2}\exp\left[-i\frac{k_\perp^2}{4A_5B_5}\right]\right\} \tag{3.121b}$$

where,

$$A_4 = \frac{\omega\Omega}{2\sin\left[\Omega(y_c^+-y^+)\right]},\qquad B_4 = \cos\left[\Omega(y_c^+-y^+)\right] \tag{3.122a}$$
$$A_5 = \frac{\omega\Omega}{2\sin\left[\Omega(L^+-y^+)\right]},\qquad B_5 = \cos\left[\Omega(L^+-y^+)\right] \tag{3.122b}$$

For the integration in the longitudinal component, the term I_4 can only be done numerically and so, for I_5 we have the following numerical expression:

$$I_5 = -\frac{4}{k_\perp^2}\mathrm{Re}\left\{1-\exp\left[-i\frac{k_\perp^2(L^+-x_0^+)\Omega}{2\omega\Omega}\right]\right\} \tag{3.123}$$

From here, it is easy to see that our total splitting function for the gluon (or more precisely, the splitting vertex) in the limit of $z \simeq 0$ is:

$$\begin{aligned}V_{g\leftarrow q}^{tot}(z\simeq 0) &= V_{g\leftarrow q}^{vac}(0)+\Delta P(0) = \\ &= 2+\frac{2}{\omega^2}\mathrm{Re}\left\{\int_{x_0^+}^{L^+} dy^+ \int_{y^+}^{L^+} dy_c^+ \left[\frac{-4A_4^2D}{(D-iA_4B_4)^2}+\frac{iA_4^3B_4k_\perp^2}{(D-iA_4B_4)^3}\right]\cdot\right.\\ &\left.\cdot\,\exp\left[-\frac{k_\perp^2}{4(D-iA_4B_4)}\right]\right\}-\frac{8}{k_\perp^2}\mathrm{Re}\left\{1-\exp\left[-i\frac{k_\perp^2(L^+-x_0^+)\Omega}{2\omega\Omega}\right]\right\}\end{aligned} \tag{3.124}$$

In the limit we are working on, the $g \to gg$ and $q \to qg$ are different only by the Casimir colour factor and so,

$$V^{tot}_{g\leftarrow g}(z\simeq 0)=V^{tot}_{g\leftarrow g}(z\simeq 1)=V^{tot}_{g\leftarrow q}(z\simeq 0) \tag{3.125}$$

Through the symmetry relations discussed in chapter 2, we are available to compute the missing splitting vertices in the considered limit:

$$V^{tot}_{q\leftarrow q}(z)=V^{tot}_{g\leftarrow q}(1-z)\Rightarrow V^{tot}_{q\leftarrow q}(z\simeq 1)=V^{tot}_{g\leftarrow q}(z\simeq 0) \tag{3.126}$$

$$V^{tot}_{q\leftarrow q}\left(\frac{1}{z}\right)=(-1)^{2s_q+2s_g-1}\frac{1}{z}V^{tot}_{q\leftarrow g}(z)\Rightarrow V^{tot}_{q\leftarrow g}(z\simeq 1)=V^{tot}_{q\leftarrow q}(z\simeq 1) \tag{3.127}$$

With these considerations, and with all the formalism of Jet-Quenching studied exhaustively in this thesis, we find the numerical expressions for the four medium-modified splitting functions, in the limit when the gluon carries a very small fraction of the energy from the projectile:

$$\begin{aligned} P^{tot}_{g\leftarrow q}(z\simeq 0)=2C_F\Bigg(1+\frac{1}{\omega^2}\mathrm{Re}\Bigg\{\int_{x_0^+}^{L^+}dy^+\int_{y^+}^{L^+}dy_c^+\left[\frac{-4A_4^2D}{(D-iA_4B_4)^2}+\frac{iA_4^3B_4k_\perp^2}{(D-iA_4B_4)^3}\right]\cdot \\ \cdot\ \exp\left[-\frac{k_\perp^2}{4(D-iA_4B_4)}\right]\Bigg\}-\frac{4}{k_\perp^2}\mathrm{Re}\left\{1-\exp\left[-i\frac{k_\perp^2(L^+-x_0^+)\Omega}{2\omega\Omega}\right]\right\}\Bigg) \end{aligned} \tag{3.128a}$$

$$\begin{aligned} P^{tot}_{q\leftarrow q}(z\simeq 1)=2C_F\Bigg(1+\frac{1}{\omega^2}\mathrm{Re}\Bigg\{\int_{x_0^+}^{L^+}dy^+\int_{y^+}^{L^+}dy_c^+\left[\frac{-4A_4^2D}{(D-iA_4B_4)^2}+\frac{iA_4^3B_4k_\perp^2}{(D-iA_4B_4)^3}\right]\cdot \\ \cdot\ \exp\left[-\frac{k_\perp^2}{4(D-iA_4B_4)}\right]\Bigg\}-\frac{4}{k_\perp^2}\mathrm{Re}\left\{1-\exp\left[-i\frac{k_\perp^2(L^+-x_0^+)\Omega}{2\omega\Omega}\right]\right\}\Bigg) \end{aligned} \tag{3.128b}$$

$$\begin{aligned} P^{tot}_{g\leftarrow g}(z\simeq 0)=2C_A\Bigg(1+\frac{1}{\omega^2}\mathrm{Re}\Bigg\{\int_{x_0^+}^{L^+}dy^+\int_{y^+}^{L^+}dy_c^+\left[\frac{-4A_4^2D}{(D-iA_4B_4)^2}+\frac{iA_4^3B_4k_\perp^2}{(D-iA_4B_4)^3}\right]\cdot \\ \cdot\ \exp\left[-\frac{k_\perp^2}{4(D-iA_4B_4)}\right]\Bigg\}-\frac{4}{k_\perp^2}\mathrm{Re}\left\{1-\exp\left[-i\frac{k_\perp^2(L^+-x_0^+)\Omega}{2\omega\Omega}\right]\right\}\Bigg) \end{aligned} \tag{3.128c}$$

$$\begin{aligned} P^{tot}_{q\leftarrow g}(z\simeq 1)=\Bigg(1+\frac{1}{\omega^2}\mathrm{Re}\Bigg\{\int_{x_0^+}^{L^+}dy^+\int_{y^+}^{L^+}dy_c^+\left[\frac{-4A_4^2D}{(D-iA_4B_4)^2}+\frac{iA_4^3B_4k_\perp^2}{(D-iA_4B_4)^3}\right]\cdot \\ \cdot\ \exp\left[-\frac{k_\perp^2}{4(D-iA_4B_4)}\right]\Bigg\}-\frac{4}{k_\perp^2}\mathrm{Re}\left\{1-\exp\left[-i\frac{k_\perp^2(L^+-x_0^+)\Omega}{2\omega\Omega}\right]\right\}\Bigg) \end{aligned} \tag{3.128d}$$

All the splitting functions computed are then used in the DGLAP evolution for the FFs. This is already implemented within a Monte Carlo program, the Q-PYTHIA [53], as well as extensions for different large z.

3.2.6 Possible Improvements in the Formalism

This formalism of medium-induced gluon radiation studied in the present thesis is used by the scientific community, as mentioned before. But, as we saw in the construction and calculation of the medium-modified splitting functions, the current state of the art formalisms needs major theoretical improvements. One of them is relax the region of validity of the spectrum, in $\mathbf{k}_\perp$ and z. Implementing this corrections in the computation of the first diagrams, will lead us to averages of at least, four Green's functions. The path integral arising from here has no known solution and so, other methods must be developed to improve the present results.

Also, in the derivation presented, we did not consider the energy loss by elastic scattering with the medium, neither intra or inter jet correlations. All these effects must be considered since they will have consequences in the hadronization process.

Colour coherence modification is another study which have not been made so far. An accurate evaluation of the corrections arising from this phenomenon, may allow us to collect all leading contributions to the spectrum.

Thus, although the formalism of Jet-Quenching is well established, a lot of scientific research is needed to contribute do the development and full comprehension of this collection of medium-induced energy loss phenomena.

3.3 Landau-Pomeranchuck-Migdal (LPM) Effect

In order to fully complete the study of the medium-induced energy loss, there is one more effect that need to be analysed: the LPM effect [54]. Like it was said in section 1.2.4, the regeneration length can have important consequences on the gluon bremsstrahlung spectrum, and so we will see the relation of the radiated energy by medium-induced gluon radiation with the formation time of the quanta itself. The LPM effect was actually was first done for QCD [55] and then specified for QED [56]. It is also responsible for the energy loss of a high energy quark (electron) or gluon due to medium stimulated gluon (photon) radiation. This effect was first considered because the corresponding problem in QCD could be a signal for QGP formation in high energy heavy-ion collisions. Nowadays, it is already implemented in simulation programs of this kind of physics [57].

Like in the previous sections, we are concerned with a quark or a gluon propagating in a medium, suffering multiple scattering and consequently emits a gluon. The difference is that now we want to know how the energy (or equivalent multiplicity) of the emitted gluon varies with its regeneration time and some medium parameters, like density and scattering centres. To do so, we will use the Gyulassy and Wang model [58, 59]. Again, the collisional energy loss will be neglected and the scattering centres are considered to be static and well described by a screening Coulomb potential. We will also assume that each scattering is independent. Beginning with one scattering, we have a total of five diagrams to the gauge invariant amplitude like shown in figure 3.22.

Before we proceed, it is useful to determine the amplitude of the elastic channel (figure 3.23) so we may remove this contribution from diagrams 3.22 to obtain simply the radiating part of the process. So, what we have is:

$$\mathcal{M}_{el} = ig^2 T^a_{AA'} T^a_{BB'} \frac{\bar{u}(p_f)\gamma_\mu u(p_i)\bar{u}(k_f)\gamma^\mu u(k_i)}{(p_i - p_f)^2 + i\epsilon} \tag{3.129}$$

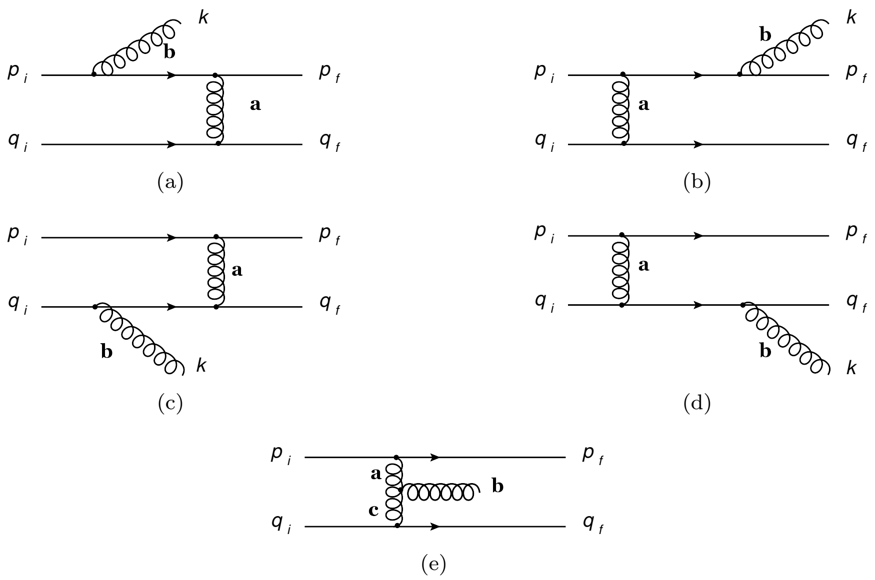


Figure 3.22: Induced gluon emission by one dispersion

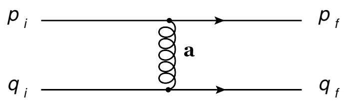


Figure 3.23: Elastic channel

The cross section is simply:

$$\frac{d\sigma}{dt} = \frac{1}{64\pi s}\frac{1}{|\mathbf{p_{1CM}}|^2}\overline{|\mathcal{M}_{el}|^2} = C_{el}\frac{4\alpha_s^2}{s^2\pi}\frac{s^2+u^2}{t^2} \tag{3.130}$$

where an average and sum over initial and final colour indices and spins is understood.

Returning to figure 3.22, if we neglect terms proportional to k_μ because we are interested in the bremsstrahlung process where the gluon is much less energetic than its parent, we have for the total amplitude:

$$\begin{aligned}
\mathcal{M}_{rad}^{(1)} = &-ig^3\frac{\bar{u}(p_f)\gamma_\mu u(p_i)\bar{u}(k_f)\gamma^\mu u(k_i)}{(k_f-k_i)^2+i\epsilon}\left\{\frac{p_f\cdot\epsilon(k)}{p_f\cdot k}T^b_{CA'}T^a_{AC}T^a_{BB'} - \frac{p_i\cdot\epsilon(k)}{p_i\cdot k}T^a_{CA'}T^b_{AC}T^a_{BB'}\right\} - \\
&- ig^3\frac{\bar{u}(p_f)\gamma_\mu u(p_i)\bar{u}(k_f)\gamma^\mu u(k_i)}{(p_f-p_i)^2+i\epsilon}\left\{T^a_{AA'}T^b_{CB'}T^a_{BC}\frac{k_f\cdot\epsilon(k)}{k\cdot k_f} - \frac{k_i\cdot\epsilon(k)}{k_i\cdot k}T^a_{AA'}T^a_{CB'}T^b_{BC}\right\} - \\
&- ig^3\frac{\bar{u}(p_f)\gamma_\mu u(p_i)\bar{u}(k_f)\gamma^\mu u(k_i)}{\left[(p_f-p_i)^2+i\epsilon\right]\left[(k_f-k_i)^2+i\epsilon\right]}T^a_{AA'}\left[T^a,T^b\right]_{BB'}\left\{\epsilon(k)\cdot(k_f-k_i)-\epsilon(k)\cdot(p_f-p_i)\right\}
\end{aligned} \tag{3.131}$$

We are free to choose the reference frame in which we want to work. So, we will choose the laboratory frame where the projectile moves very fast in the $+\hat{z}$ direction, and give to the target thermal freedom movement, M. Our kinematic is (the gauge is still $A^+ = 0$):

$$p_i = \left(P^+,\ 0,\ \mathbf{0}\right) \tag{3.132a}$$
$$k_i = (M,\ M,\ \mathbf{0}) \tag{3.132b}$$
$$q = \left(q^+,\ q^-,\ \mathbf{q}_\perp\right) \tag{3.132c}$$
$$k = \left(xP^+,\ \frac{\mathbf{k}_\perp{}^2}{2xP^+},\ \mathbf{k}_\perp\right) \tag{3.132d}$$
$$\epsilon = \left(0,\ \frac{\boldsymbol{\epsilon}_\perp \cdot \mathbf{k}_\perp}{xP^+},\ \boldsymbol{\epsilon}_\perp\right) \tag{3.132e}$$

What we are trying to prove now is that in the chosen reference frame, the contribution from diagrams 3.22(c) and 3.22(d) can be neglected when compared to diagrams 3.22(a) and 3.22(b) since the structure is almost the same. Now, for the final 4-vectors, we have two choices depending on the diagram we are analyzing:

$$p_f = p_i - q - k \tag{3.133a}$$
$$k_f = k_i + q \tag{3.133b}$$

or,

$$p_f = p_i - q \tag{3.134a}$$
$$k_f = k_i + q - k \tag{3.134b}$$

But in order to fully specify them, we have to determine the q-components through conditions $p_f^2 = k_f^2 = q^2 = 0$. Considering the limit $x \ll 1$ but high enough so that $xP^+ \gg M \gg q_\perp$, we can have for the configuration (3.133):

$$q^- \simeq -\frac{\mathbf{k}_\perp \cdot \mathbf{q}_\perp}{(1-x)P^+} - \frac{k_\perp^2}{2xP^+} \tag{3.135a}$$
$$q^+ \simeq \frac{\mathbf{k}_\perp \cdot \mathbf{q}_\perp}{(1-x)P^+} + \frac{k_\perp^2}{2xP^+} - M \tag{3.135b}$$

and finally arrive at:

$$\frac{\epsilon \cdot p_f}{k \cdot p_f} \approx \frac{\epsilon \cdot p_i}{k \cdot p_i} = 2\frac{\boldsymbol{\epsilon}_\perp \cdot \mathbf{k}_\perp}{k_\perp^2} \tag{3.136}$$

$$\frac{\epsilon \cdot k_i}{k \cdot k_i} = \frac{\boldsymbol{\epsilon}_\perp \cdot \mathbf{k}_\perp}{(xP^+)^2} \tag{3.137}$$

$$\frac{\epsilon \cdot k_f}{k \cdot k_f} = -\frac{\boldsymbol{\epsilon}_\perp \cdot \mathbf{q}_\perp}{xP^+M} \tag{3.138}$$

From section 1.2.4, we saw that there are three cases which allow to have a large total emission probability. In all of them, $k_\perp \ll k = xP^+$ or $k_\perp \sim xP^+$. So, the contribution of the diagrams in which the target radiate can be neglected. Of course we can only do this by means of energy counting. Continuing, we have as our new amplitude:

$$\mathcal{M}_{rad}^{(1)} = -g\frac{\mathcal{M}_{el}}{T^a_{AA'}T^a_{BB'}}2\boldsymbol{\epsilon}_\perp\left\{\frac{\mathbf{k}_\perp}{k_\perp^2} - \frac{(\mathbf{k}_\perp - \mathbf{q}_\perp)}{(\mathbf{k}_\perp - \mathbf{q}_\perp)^2}\right\}\left[T^a, T^b\right]_{AA'}T^a_{BB'} \equiv i\frac{\mathcal{M}_{el}\mathcal{R}_1}{T^a_{AA'}T^b_{BB'}} \tag{3.139}$$

$$\mathcal{R}_1 = 2ig\boldsymbol{\epsilon}_\perp \cdot \left\{\frac{\mathbf{k}_\perp}{k_\perp^2} - \frac{(\mathbf{k}_\perp - \mathbf{q}_\perp)}{(\mathbf{k}_\perp - \mathbf{q}_\perp)^2}\right\}\left[T^a, T^b\right]_{AA'}T^a_{BB'} \tag{3.140}$$

with $\mathcal{R}_1$ defined as the radiation amplitude induced by a single scattering and where we have used $T^a_{AA'}\left[T^a, T^b\right]_{BB'} = -\left[T^a, T^b\right]_{AA'}T^a_{BB'}$. Making use of the Gunion-Bertsch formula [60], we can immediately see how the gluon spectrum looks like:

$$\frac{dn^{(1)}}{dy\, d^2k_\perp} \equiv \frac{\overline{|\mathcal{R}_1|}}{2(2\pi)^2 C_{el}} = \frac{C_A\alpha_s}{\pi^2}\frac{q_\perp^2}{(\mathbf{k}_\perp - \mathbf{q}_\perp)^2 k_\perp^2} \tag{3.141}$$

This expression has two singularities: the well known infrared singularity $\mathbf{k}_\perp = 0$ and the singularity $\mathbf{k}_\perp = \mathbf{q}_\perp$ which comes exclusively from the non-Abelian nature of the theory since it appears due diagram 3.22(e). If $\mathbf{k}_\perp \ll \mathbf{q}_\perp$, then this contribution can also be neglected when compared to the leading contribution $1/k_\perp^2$. If not, this contribution is important since it changes the behavior of the gluon spectrum to $1/k_\perp^4$. Therefore, we can say $q_\perp$ act as a cut-off for $k_\perp$ when one neglects the amplitude with the three-gluon vertex, as we will do in the next.

This means the radiation amplitude can be written as:

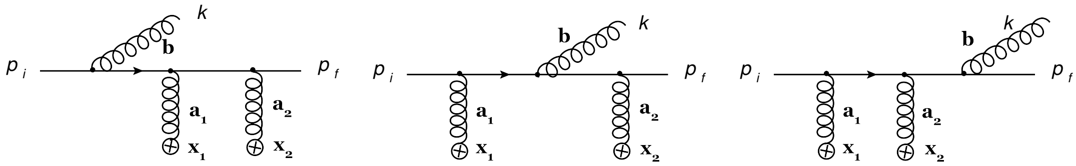
$$\mathcal{R}_1 = 2ig\frac{\boldsymbol{\epsilon}_\perp \cdot \mathbf{k}_\perp}{k_\perp^2}\left[T^a, T^b\right]_{AA'}T^a_{BB'}\mathrm{e}^{ik\cdot x_1} \tag{3.142}$$

being x_1 the target parton position.

For two scatterings like in figure 3.24 we have:

$$\mathcal{R}_2 = 2ig\frac{\boldsymbol{\epsilon}_\perp \cdot \mathbf{k}_\perp}{k_\perp^2}\left\{\left(T^{a_2}\left[T^{a_1}, T^b\right]\right)_{AA'}\mathrm{e}^{ik\cdot x_1} + \left(\left[T^{a_2}, T^b\right]T^{a_1}\right)_{AA'}\mathrm{e}^{ik\cdot x_2}\right\}\left(T^{a_1}T^{a_2}\right)_{BB'} \tag{3.143}$$

and extrapolating to m scatterings:



(a) (b) (c)

Figure 3.24: Induced gluon emission by two dispersions

$$\mathcal{R}_m = 2g\frac{\boldsymbol{\epsilon}_\perp \cdot \mathbf{k}_\perp}{k_\perp^2}\left\{\sum_{i=1}^{m}\left(T^{a_m}\cdots\left[T^{a_i},T^b\right]\cdots T^{a_1}\right)_{AA'}\mathrm{e}^{ik\cdot x_i}\right\}\left(\prod_{i=1}^{m}T^{a_1}\right)_{BB'} \tag{3.144}$$

one gets the high energy limit radiation amplitude, which is also valid for a gluon beam if the colour matrices are written in the adjoint representation. With this, the spectrum for the bremsstrahlung process associated with multiple scattering in colourless objects is:

$$\frac{dn^{(m)}}{dyd^2k_\perp} = \frac{1}{2(2\pi)^3 C_{el}^m}\overline{|\mathcal{R}_m|^2} \equiv C_m(k)\frac{dn^{(1)}}{dyd^2k_\perp} \tag{3.145}$$

where $C_m(k)$ is defined as the *radiation formation factor* to characterize the interference pattern due to multiple scattering. It can be expressed as:

$$C_m(k) = \sum_{i=1}^{m}\left\{1 - \mathrm{Re}\sum_{j=1}^{i-1} r_2(1-r_2)^{i-j-1}\mathrm{e}^{ik\cdot(x_i-x_j)}\right\} \tag{3.146}$$

with the parameter $r_2 = C_A/(2C_F)$. In this way, it is possible to get a generic expression for a high energy parton jet if:

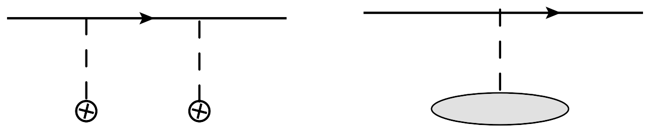
$$r_2 = \frac{C_A}{2C_2} = \begin{cases} \frac{C_A}{2C_F} = \frac{N_c^2}{N_c^2-1} & \text{for quarks } (C_2 = C_2(N) = C_F) \\ \frac{C_A}{2C_A} = \frac{1}{2} & \text{for gluons } (C_2 = C_2(G) = C_A) \end{cases} \tag{3.147}$$

First of all, lets note that in the case of one scattering ($m = 1$) we recover the gluon spectrum induced by single scattering in the small $\mathbf{k}_\perp$ limit because $C_1(k) = 1$. As for the phase factor $k \cdot (x_i - x_j)$, it determines the interference between radiation from two successive scatterings (i, j) and is simply the ratio of the path length L to the formation time:

$$k \cdot (x_i - x_j) \equiv \frac{L}{\tau(k)} \tag{3.148}$$

In the limit $L \gg \tau(k)$, the induced radiation intensity is additive in the number os scatterings because the phase factor can be averaged to zero. This is called the Bethe-Heitler limit where $C_m(k) \approx m$: scattering centres act like independent sources of radiation since the field formation time is short enough to regenerate itself before a new interaction. Thus, for each interaction there is a new exchange of information (see fig 3.25(a)).

However, when $L \ll \tau(k)$, the final state radiation amplitude from the first scattering is mostly



(a) Limit of Bethe-Heitler

(b) Limit of factorization or LPM effect

canceled by the initial state radiation amplitude from the second scattering. The radiation pattern then looks as if the parton only suffered a single scattering from p_i to p_f (see figure 3.25(b)). This destructive interference is referred to as LPM effect and the corresponding limit is usually called factorization limit. In this case the phase factor can be set to unity and the radiation formation factor simplified:

$$C_m(k) \approx \frac{1}{r_2}\left[1-(1-r_2)^m\right] = \begin{cases} 8/9\left[1-(-1/8)^m\right] & \text{for quarks} \\ 2\left[1-(1/2)^m\right] & \text{for gluons} \end{cases} \tag{3.149}$$

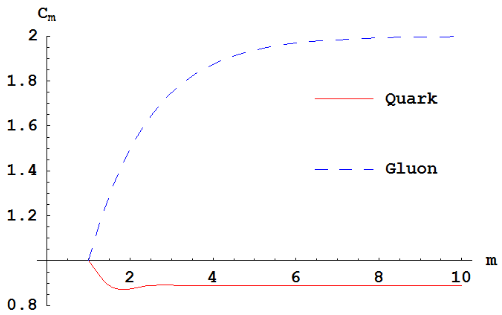


Figure 3.25: Radiation formation factor for QCD: C_m vs m

The graphic results are shown in figure 3.25. We already saw what is the spectrum of the radiated gluon if the parent parton suffers one scattering. Now, we are analyzing how much does it varies if we increase the number os scattering centres. Somehow, a more natural expectation was the Bethe-Heitler limit because if the parton interacts twice with the medium, it could loss (nearly) twice the energy from one scattering. But, because of the interference mentioned before, the parton does not lose so many energy as expected and the growth of the radiation formation factor starts to decrease until the value $1/r_2$. For gluons, this value is still higher then $C_1(k)$, but for quarks, $C_\infty < C_1(k)$. This interference is so effective that the radiation spectrum induced by many scatterings is even less than by a single scattering and the parent parton does not loose so many energy. It is curious to see that a single scattering is more effective than, for example, two scatterings. This graphic also show us that a gluon radiates more than a quark, which is natural to think since it possesses more colour charge.

Now we can discuss how the effective formation time of the radiation in a QCD medium can affect the energy loss. For that and to see analytically how $C_m(k)$ interpolates between the two limits considered, we may average over the interaction points $\mathbf{x_i}$ according to a linear kinetic theory [61]. This tell us that the longitudinal separation between two successive scattering centres, $L_i = z_{i+1} - z_i$, has a distribution controlled by the mean free path, λ.

$$\frac{dP}{dL_i} = \frac{1}{\lambda}e^{-L_i/\lambda} \tag{3.150}$$

The averaging over the phase factors is then:

$$\left\langle e^{ik\cdot(x_i-x_j)}\right\rangle \approx \left[\frac{1}{1-i\lambda/\tau(k)}\right]^{i-j} \equiv \left[\frac{1}{1-i\chi(k)}\right]^{i-j} \tag{3.151}$$

which enable us to come to the following expression:

$$C_m(k) \approx m\frac{(\chi(k)/r_2)^2}{1+(\chi(k)/r_2)^2} + \frac{1-(1-2r_2)(\chi(k)/r_2)^2}{r_2\left(1+(\chi(k)/r_2)^2\right)} \tag{3.152}$$

if we use the series sum:

$$\sum_{i=1}^{m}\sum_{j=1}^{i-1} x^{i-j} = \frac{x(x^m-1)-mx(x-1)}{(x-1)^2} \tag{3.153}$$

and neglect terms proportional to $(1-r_2)^m$, an approximation only valid for a relative high m. To see the behavior of expression 3.152, we have to fix the value of m. The results are displayed in figure 3.26 for $m = 10$.

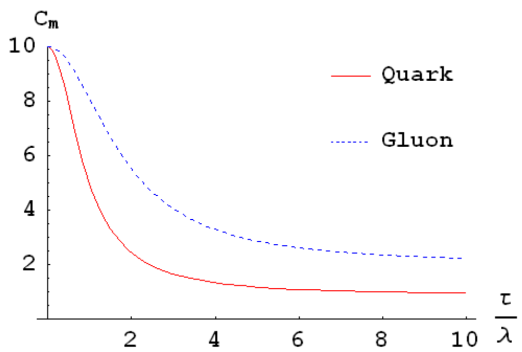
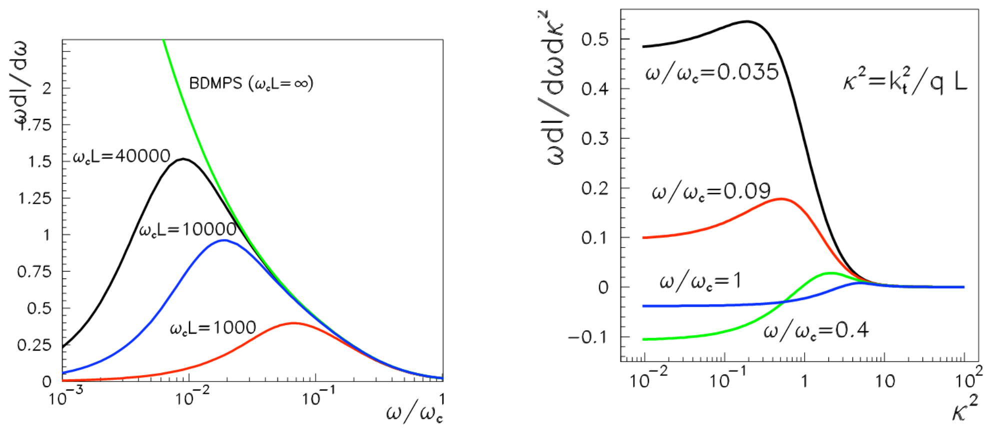


Figure 3.26: Radiation formation factor for QCD: C_m vs τ/λ

Now we can see easily that when the regeneration time is small enough to resolve all scattering centres in the medium we have the Bethe-Heitler limit. As τ goes increasing, the projectile cannot *observe* the environment well enough and instead of m scattering centres, it exchange information with a diffuse block of matter (fig. 3.25(b)) and consequently it does not loose so much energy. Soon, the value of $C_m(k)$ decreases and tends once again to stabilize and saturate to the limit of factorization. It is also visible the difference between quarks and gluons.

3.3.1 The LPM effect and the Spectrum of Radiated Gluons

The LPM effect enable us to explain the supression of the spectrum (3.108) for large values of ω (fig. 3.27(a)) and for small values of $k_\perp$ (fig. 3.27(b)).



(a) (b)

Figure 3.27: Numerical results for the medium induced gluon radiation spectrum (3.108)

The results from figure 3.27 are given as a function of dimensionless variables:

$$\omega_c \equiv \frac{1}{2}\hat{q}L^2 \tag{3.154a}$$

$$\kappa^2 \equiv \frac{k_\perp^2}{\hat{q}L} \tag{3.154b}$$

The formation time can be computed from the non-eikonal contributions performed in previous sections:

$$\prod \mathrm{e}^{i\frac{k_\perp^2}{2p^+}(x_i - x_{i-1})^+} \simeq \mathrm{e}^{i\frac{k_\perp^2}{2\omega}L} = \mathrm{e}^{i\frac{L}{t_{form}}} \tag{3.155}$$

where,

$$t_{form} = \frac{2\omega}{k_\perp^2} \tag{3.156}$$

This means, that for large values of ω or small values of $k_\perp$, the formation time is much larger than the medium size and the projectile cannot exchange information with all scattering centres. The whole medium acts as a single scattering centre as a reduction of the spectrum is produced.

Chapter 4

Summary and Conclusions

The theory of QCD and the study of Jet-Quenching were the main subjects of this thesis.

We saw briefly the parton model and the immediate consequent of Bjorken scaling, which tell us that the evolution of the PDFs depends only on the Bjorken-x variable. This is only an approximation and deviations with $\log(Q^2)$ are expected to occur. These violations comes from the fact that QCD is a non-Abelian theory and although the coupling constant is small for large values of Q^2, its result is different from zero. In this regime, perturbative QCD is a powerful and very precise method capable of giving approximate analytic solutions through a perturbative expansion. On the other hand, when the distance between two partons increases, the coupling constant also increase and so, pQCD is no longer valid in this regime. The scale that separates these two regimes is the theory scale, Λ_{QCD}.

Besides the perturbation theory, factorization is another mechanism that makes calculation of experimental observables possible in QCD. It consists of separating a collision between two hadrons into three distinct parts: a non-perturbative evolution of both hadrons until the moment of interaction (the PDFs); the moment of the collision between the constituents partons during which there is exchange of momentum (this is the perturbative cross section); and the subsequent process where the partons ejected from hadrons hadronize through non-perturbative processes (FFs).

PDFs and FFs are universal and dynamical evolution quantities. Its evolution is governed by distinct evolution equations, depending on what type of evolution we are making in the phase space (Q^2, x). In this thesis, we worked in very high energies where saturation can be neglected. The equations that describe this kind of evolution are the DGLAP equations, a set of coupled differential equations for the FFs or PDFs. It shows that any attempt to measure one individual parton, will result in a split of this parton into two new particles, each with a fraction z and $(1-z)$ from the initial one. This is controlled by the four splitting functions: quark to quark $(q \leftarrow q)$, quark to gluon $(g \leftarrow q)$, gluon to gluon $(g \leftarrow g)$ and gluon to quark $(q \leftarrow g)$. This will result in a decrease with Q^2 of the PDFs (or FFs) to higher values of x and a increase for smaller values of x.

But this behavior might be very different when these kind of processes take place inside a medium. There will be more interactions resulting in an additional energy loss. This is supposed to happen in AA collisions since the colliding objects have spatial extension. Depending on where the collision takes place (near the border of the system or deep inside) the resulting jet might be quenched. The collection of phenomena responsible for this medium-induced energy loss is called Jet-Quenching.

There are many models that try to explain this phenomenon and get quantitative predictions for some of the medium parameters that characterize the QGP formed by the collision. The differences are essentially about their assumptions since there is need to make simplifications in order to make some kind of calculations. We followed the BDMPS-Z/ASW approach, where the medium, considered as a static background, is characterized by an ensemble of independent scattering centres. The parton suffers multiple soft scattering within the medium and escape with a smaller energy.

With these considerations, we computed the amplitude for a quark propagating eikonally inside the medium, deriving in this way the Wilson line. Next we relax this assumption by permitting some Brownian motion in the transverse plane, obtaining in this way the Green's function. We applied this to our study of a quark transversing a medium and emitting a gluon, in the limit that the gluon carries very little of the initial energy.

Because we considered that the time of propagation through the target is very small, we assumed that the gluonic field does not vary during the propagation. So, for the cross section, we had to average over the fields that characterize the target. Some formulas were derived and applied to the computation of our spectrum.

To get a numerical result for the medium-modified splitting functions, we had to assume that the total function could be written as a sum of a medium and vacuum part. Furthermore, we assumed that the medium splitting function could be taken from the spectrum by comparison with the vacuum one. To solve the path-integral, we used the dipole approximation resulting in an expression equivalent to that of a harmonic oscillator.

With all these, we obtained a total of three terms: a medium-independent contribution, which gives the vacuum splitting function; and two more medium-dependent contributions, an additive term (direct production) and a destructive interference term. These corrections are present in all splitting functions in the limit considered and depends on the transverse momentum and energy of the emitted parton. The destructive interference term, is responsible for making the spectrum finite for small values of ω, like initially predicted by the BDMPS formalism, since the only diagram computed was the one in which the gluon is prodiced inside the medium. So, instead of having a spectrum that diverges to infinity, we have a finite result.

In the oposite limit (large values of ω) and for small values of $k_\perp$ is observed a supression of the spectrum. The physical effect behind this phenomenon is the LPM effect. This occurs essentially due the finiteness of the regeneration time of the proper field of the propagating parton. If it is greater than the distance of two successive scattering centres, the projectile sees actually a diffuse block of matter and only interacts once. So, the energy loss is less that what is should be expected.

All the splitting functions are already implemented in Monte Carlo programs, like Q-PYTHIA.

Appendix A

Light-Cone Conventions

In special relativity, a light cone is the surface describing the temporal evolution of a flash of light in Minkowski space. It can be visualized in figure A.1.

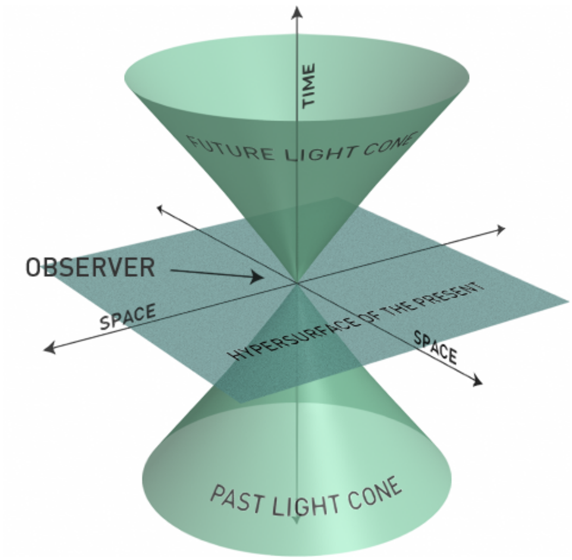


Figure A.1: Representation of a light cone

There is two main conventions for the light-cone quantization: the Lepage-Brodsky (LP) representation [62–65] which essentially works with the Dirac representation for the Dirac matrices but has a definition of 4-vectors components which requires two different metrics $g_{\mu\nu}$ and $g^{\mu\nu}$; and the Kogut-Soper (KS) convention [65,66] where $g_{\mu\nu} = g^{\mu\nu}$ but works with the chiral representation for the Dirac matrices instead.

Because I prefer to work with the Dirac representation rather than the chiral one, I will join the best of the two conventions. So, a contra-variant Lorentz vector is given by:

$$x^{\mu} = (x^{+},\ x^{-},\ \mathbf{x}_{\perp}) \tag{A.1}$$

where the relation with the usual Minkowsky coordinates is as follows:

$$x^+ = \frac{x^0 + x^3}{\sqrt{2}} \,, \qquad x^- = \frac{x^0 - x^3}{\sqrt{2}} \,, \qquad \mathbf{x}_\perp = \left(x^1,\; x^2\right) \tag{A.2}$$

To the coordinate x^+ it is called the light-cone time and to x^- the light-cone longitudinal position. $\mathbf{x}_\perp$ are the transverse coordinates. The metric chosen is:

$$g_{\mu\nu} = g^{\mu\nu} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \tag{A.3}$$

which means that a covariant 4-vector is given by:

$$x_\mu = g_{\mu\nu} x^\mu = (x_+,\; x_-,\; \mathbf{x}_\perp^{'}) = (x^-,\; x^+,\; -\mathbf{x}_\perp) \tag{A.4}$$

So, the scalar product between the two 4-vectors is:

$$x \cdot p = x^\mu p_\mu = g_{\mu\nu} x^\mu p^\nu = x^+ p^- + x^- p^+ - \mathbf{x}_\perp \cdot \mathbf{p}_\perp \tag{A.5}$$

Remembering the Dirac representation:

$$\gamma^0 = \beta = \begin{pmatrix} \mathbf{1} & 0 \\ 0 & -\mathbf{1} \end{pmatrix} \quad , \quad \boldsymbol{\gamma} = \beta \boldsymbol{\alpha} = \begin{pmatrix} 0 & \boldsymbol{\sigma} \\ -\boldsymbol{\sigma} & 0 \end{pmatrix} \tag{A.6}$$

we can apply this formalism to the Dirac matrices and obtain:

$$\gamma^+ = \frac{\gamma^0 + \gamma^3}{\sqrt{2}} = \frac{1}{\sqrt{2}} \begin{pmatrix} \mathbf{1} & \sigma^3 \\ -\sigma^3 & \mathbf{1} \end{pmatrix} \tag{A.7a}$$

$$\gamma^- = \frac{\gamma^0 - \gamma^3}{\sqrt{2}} = \frac{1}{\sqrt{2}} \begin{pmatrix} \mathbf{1} & -\sigma^3 \\ \sigma^3 & \mathbf{1} \end{pmatrix} \tag{A.7b}$$

$$\gamma^1 = \begin{pmatrix} 0 & \sigma^1 \\ -\sigma^1 & 0 \end{pmatrix} \tag{A.7c}$$

$$\gamma^2 = \begin{pmatrix} 0 & \sigma^2 \\ -\sigma^2 & 0 \end{pmatrix} \tag{A.7d}$$

where $\mathbf{1}$ is the identity matrix and σ^i the Pauli matrices:

$$\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad , \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad , \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{A.8}$$

These new matrices obey to the following relations:

$$\gamma^+\gamma^+ = \gamma^-\gamma^- = 0 \tag{A.9a}$$

$$\gamma^+\gamma^-\gamma^+ = 2\gamma^+ \tag{A.9b}$$

$$\gamma^-\gamma^+\gamma^- = 2\gamma^- \tag{A.9c}$$

Spinors of momentum p and spin λ are now given by:

$$u(p,\ \lambda) = \frac{1}{\sqrt{\sqrt{2}p^+}}\left(\sqrt{2}p^+ + \beta m + \boldsymbol{\alpha}_\perp \cdot p_\perp\right)\begin{cases} \chi(\uparrow), \text{if } \lambda = +1 \\ \chi(\downarrow), \text{if } \lambda = -1 \end{cases} \tag{A.10}$$

$$v(p,\ \lambda) = \frac{1}{\sqrt{\sqrt{2}p^+}}\left(\sqrt{2}p^+ - \beta m + \boldsymbol{\alpha}_\perp \cdot p_\perp\right)\begin{cases} \chi(\downarrow), \text{if } \lambda = +1 \\ \chi(\uparrow), \text{if } \lambda = -1 \end{cases} \tag{A.11}$$

or, explicitly:

$$u(p,\ \lambda) = \frac{1}{\sqrt{\sqrt{2}p^+}}\left[\begin{array}{cc} \begin{pmatrix} \sqrt{2}p^+ + m & 0 \\ 0 & \sqrt{2}p^+ + m \end{pmatrix} & \begin{pmatrix} 0 & p_1 - ip_2 \\ p_1 + ip_2 & 0 \end{pmatrix} \\ \begin{pmatrix} 0 & p_1 - ip_2 \\ p_1 + ip_2 & 0 \end{pmatrix} & \begin{pmatrix} \sqrt{2}p^+ + m & 0 \\ 0 & \sqrt{2}p^+ + m \end{pmatrix} \end{array}\right]\begin{cases} \chi(\downarrow),\ \lambda = +1 \\ \chi(\uparrow),\ \lambda = -1 \end{cases} \tag{A.12}$$

where the two χ-spinors are:

$$\chi(\uparrow) = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} \quad , \quad \chi(\downarrow) = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix} \tag{A.13}$$

or generally speaking:

$$\chi(\lambda \uparrow) = \frac{1}{\sqrt{2}}\begin{pmatrix} \phi(\lambda \uparrow) \\ \lambda\phi(\lambda \uparrow) \end{pmatrix} \tag{A.14}$$

where:

$$\phi(+\uparrow) = \phi(\uparrow) = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad \phi(-\uparrow) = \phi(\downarrow) = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \tag{A.15}$$

These elements must obey to the following relations:

$$(\not{p} + m)u(p) = 0 \tag{A.16}$$
$$(\chi_\lambda)^\dagger \chi_{\lambda'} = \delta_{\lambda\lambda'} \tag{A.17}$$

In order to specify the polarization vector, a gauge must be fixed. For that, we have to obey to [7,67]:

$$\partial_\mu A^\mu = 0 \Leftrightarrow p_\mu A^\mu = 0 \Leftrightarrow p^+ A^- + p^- A^+ - \mathbf{p}_\perp \cdot \mathbf{A}_\perp = 0 \tag{A.18}$$

In the high energy limit, a particle propagating in the $+\hat{z}$ direction has the component $p^- = 0$. So, the previous condition is now of the form:

$$p^+ A^- - \mathbf{p}_\perp \cdot \mathbf{A}_\perp = 0 \tag{A.19}$$

which means the only choice we have is to put $A^+ = 0$. For this condition, the polarization vector is of the form:

$$\epsilon^\mu(p,\ \lambda) = \left(0,\ \frac{\boldsymbol{\epsilon}_\perp \cdot p_\perp}{p^+},\ \epsilon_\perp(\lambda)\right) \tag{A.20}$$

with

$$\boldsymbol{\epsilon}_\perp(\lambda) = -\frac{1}{\sqrt{2}}\left(\lambda \mathbf{e_x} + i\mathbf{e_y}\right) \tag{A.21}$$

It must obey to:

$$\epsilon^{*\mu}(p,\ \lambda)\epsilon_\mu(p,\ \lambda') = -\delta_{\lambda\lambda'} \tag{A.22}$$
$$p^\mu \epsilon_\mu(p,\ \lambda) = 0 \tag{A.23}$$
$$d_{\mu\nu} = \sum_\lambda \epsilon_\mu(p,\lambda)\epsilon^*_\nu(p,\ \lambda) = -g_{\mu\nu} + \frac{\eta_\mu p_\nu + \eta_\nu p_\mu}{p^k \eta_k} \tag{A.24}$$

where η^μ is the null vector:

$$\eta^\mu = (0,\ 1,\ \mathbf{0}) \quad , \quad \eta^\mu \eta_\mu = 0 \tag{A.25}$$

Appendix B

Feynman Rules for a non-Abelian Gauge Theory

For a non-Abelian gauge theory like QCD, the usual Feynam rules for QED are slightly altered [7, 15, 49]. It must have the introduction of colored matrices T^a, the generators of the group $SU(N)$, the exchange of $-e$ by g and the addition of new vertices due the self-interacting gauge boson nature. Some of these rules are listed bellow and one must pay attention to the momentum direction in each diagram.

Feynman Rules for Internal Lines:

- Incoming fermion: $u(p)$
- Outcoming fermion: $\bar{u}(p)$
- Incoming anti-fermion: $\bar{v}(p)$
- Outcoming anti-fermion: $v(p)$
- Incoming boson: $\epsilon^{\mu}(k)$
- Outcoming boson: $\epsilon^{*\mu}(k)$

Feynman Rules for External Lines:

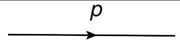
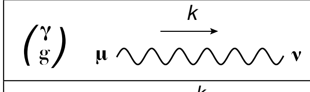
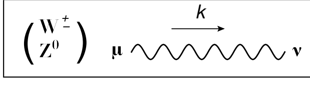
Diagram	Analytical Expression
	$\frac{i(\not{p}+m)}{p^2-m^2+i\epsilon}$
	$\frac{-ig_{\mu\nu}}{k^2+i\epsilon}$
	$\frac{-ig_{\mu\nu}-k_\mu k_\nu/M^2}{k^2+i\epsilon}$

Table B.1: Feynam Rules for External Lines

Feynman Rules for Vertices

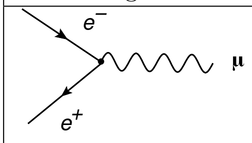
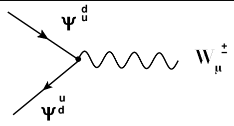
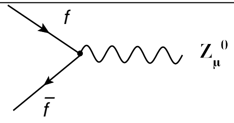
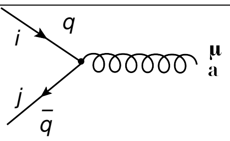
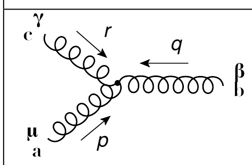
Diagram	Analytical Expression
	$-ie\gamma^\mu$
	$\frac{ig}{\sqrt{2}}\gamma^\mu\left(\frac{1-\gamma^5}{2}\right)$
	$\frac{ig}{\cos\theta_w}\gamma^\mu\left(g^f_v - g^f_A\gamma^5\right)$
	$ig\gamma^\mu T^a_{ji}$
	$V^{abc}_{\alpha\beta\gamma} = gf^{abc}\left[g_{\alpha\beta}(p-q)_\gamma + g_{\beta\gamma}(q-r)_\alpha + g_{\gamma\alpha}(r-p)_\beta\right]$

Table B.2: Feynam Rules for Vertices

Appendix C

Colour Algebra

The generators of the symmetry group T^a are traceless and must obey to a Lie Algebra:

$$\left[T^a, T^b\right] = if^{abc}T^c \tag{C.1}$$

where f^{abc} is a set of numbers called structure constants which obey to Jacobi identity:

$$\begin{gathered}\left[T^a, \left[T^b, T^c\right]\right] + \left[T^b, \left[T^c, T^a\right]\right] + \left[T^c, \left[T^a, T^b\right]\right] = 0 \Leftrightarrow \\ \Leftrightarrow f^{ade}f^{bcd} + f^{bde}f^{cad} + f^{cde}f^{abd} = 0\end{gathered} \tag{C.2}$$

Each group must have representations which may be found to be irreducible, T_r. Let $r = N$ be the basic or fundamental irreducible representation (quarks) and $\bar{r} = \bar{N}$ the conjugate representation (anti-quarks). So, the dimension of these representations in the $SU(N)$ group is $d(r) = d(\bar{r}) = N_c$ (number of colors). Another irreducible representation is the one to which the generators of the algebra belong. This is designated by adjoint representation, $r = G$ and corresponds to the eight gluons:

$$(T^a_G)_{bc} = -if^{abc} \tag{C.3}$$

$$d(G) = N_c^2 - 1 \tag{C.4}$$

In general, we have

$$\text{Tr}\left[T^a_r T^b_r\right] = C(r)\delta^{ab} \tag{C.5}$$

where $C(r)$ is a constant for each representation called Casimir. We have:

$$C(N) = \frac{1}{2} \quad \text{and} \quad C(G) = N_c \tag{C.6}$$

It can be shown that $T^2 = T^a T^a$ (as usual sum over repeated indices) is an invariant of the algebra since it commutes with all groups generators. Thus, the matrix representation of T^2 is proportional to the unitary matrix:

$$t_r^a t_r^a = C_2(r) \cdot \mathbf{1} \tag{C.7}$$

where $\mathbf{1}$ is the $d(r) \times d(r)$ unit matrix and $C_2(r)$ a constant for each representation called the quadratic Casimir operator. For the adjoint representation:

$$f^{acd} f^{bcd} = C_2(G) \delta^{ab} \tag{C.8}$$

and so we can relate both Casimir by the simple expression

$$d(r) C_2(r) = d(G) C(r) \tag{C.9}$$

which means that $C_2(N) \equiv C_F = (N_c^2 - 1)/(2N_c)$ and $C_2(G) \equiv C_A = N_c$.

Another important relation is the Fierz Identity:

$$T_{ij}^a T_{kl}^a = \frac{1}{2} \left(\delta_{il} \delta_{jk} - \frac{1}{N_c} \delta_{ij} \delta_{kl} \right) \tag{C.10}$$

This, together with previous relations, allow us to prove several useful relations:

$$T_{ij}^a T_{jk}^a = C_F \delta_{ik} \tag{C.11}$$

$$f^{abc} f^{*abd} = C_A \delta^{cd} \tag{C.12}$$

$$\mathrm{Tr}\left[T^a T^a\right] = \frac{N_c^2 - 1}{2} = N_c C_F \tag{C.13}$$

$$\mathrm{Tr}\left[T^a T^a T^b\right] = 0 \tag{C.14}$$

$$\mathrm{Tr}\left[T^a T^a T^b T^c\right] = C_F \mathrm{Tr}\left[T^b T^c\right] \tag{C.15}$$

$$\mathrm{Tr}\left[T^a T^b T^a T^c\right] = -\frac{1}{2N_c} \mathrm{Tr}\left[T^b T^c\right] \tag{C.16}$$

$$\mathrm{Tr}\left[T^a T^a T^b T^c T^d\right] = C_F \mathrm{Tr}\left[T^b T^c T^d\right] \tag{C.17}$$

$$\mathrm{Tr}\left[T^a T^b T^a T^c T^d\right] = -\frac{1}{2N_c} \mathrm{Tr}\left[T^b T^c T^d\right] \tag{C.18}$$

$$\mathrm{Tr}\left[T^a T^a T^b T^c T^d T^e\right] = C_F \mathrm{Tr}\left[T^b T^c T^d T^e\right] \tag{C.19}$$

$$\mathrm{Tr}\left[T^a T^b T^a T^c T^d T^e\right] = -\frac{1}{2N_c} \mathrm{Tr}\left[T^b T^c T^d T^e\right] \tag{C.20}$$

$$\mathrm{Tr}\left[T^a T^b T^c T^a T^d T^e\right] = \frac{1}{2} \mathrm{Tr}\left[T^b T^c\right] \mathrm{Tr}\left[T^d T^e\right] - \frac{1}{2N_c} \mathrm{Tr}\left[T^b T^c T^d T^e\right] \tag{C.21}$$

$$\mathrm{Tr}[T^a T^b T^c] \mathrm{Tr}\left[T^c T^d T^e\right] = \frac{1}{2} \mathrm{Tr}\left[T^a T^b T^e T^d\right] - \frac{1}{2N_c} \mathrm{Tr}\left[T^a T^b\right] \mathrm{Tr}\left[T^d T^e\right] \tag{C.22}$$

$$\mathrm{Tr}\left[T^aT^bT^cT^d\right]\mathrm{Tr}\left[T^dT^eT^fT^g\right]=\frac{1}{2}\mathrm{Tr}\left[T^aT^bT^cT^eT^fT^g\right]-\frac{1}{2N_c}\mathrm{Tr}\left[T^aT^bT^c\right]\mathrm{Tr}\left[T^eT^fT^g\right] \quad \text{(C.23)}$$

$$\mathrm{Tr}\left[T^aT^bT^cT^d\right]\mathrm{Tr}\left[T^dT^cT^bT^a\right]=\frac{N_cC_F^3}{2}-\frac{C_F^2}{4}+\frac{C_F}{8N_c} \quad \text{(C.24)}$$

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